Lecture notes for May 6 to June 11

Chengjian Yao, ShanghaiTech University

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1 Line integral

1.1 Line integral of scalar fields

Let $L \subset \mathbf{R}^3$ be a curve and f be a scalar function on it. We can consider the integral

$$\int_L f d\sigma_1 = \int_L f ds \tag{1.1}$$

using the notion of Riemann sum $\lim_{\|T\|\to 0} \sum_{i=1}^n f(P_i^*)\Delta s_i$, where Δs_i is the length of the i-th arc under the partition T.

In case $f \equiv 1$, the integral represents the total length of the curve. In physical application, if f represents the line density of a very thin wire, then the integral represents the total mass of the wire.

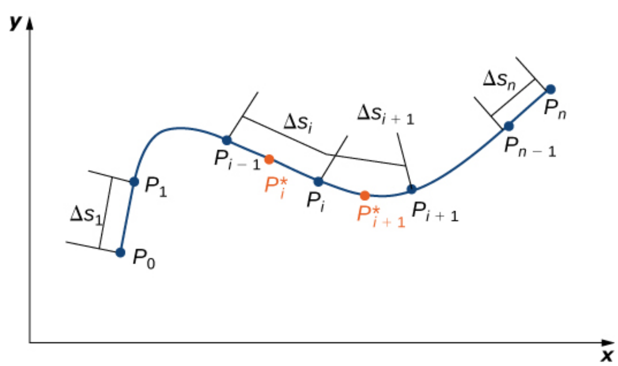


Figure 1: Line integral of a scalar function

If $\boldsymbol{r} : [\alpha, \beta] \to \mathbf{R}^3$ is a C^1 parametrization of the curve, then

$$ds = |\boldsymbol{r}'(t)|\, dt,$$

and therefore

$$\int_L f ds = \int_\alpha^\beta f(\boldsymbol{r}(t))\, |\boldsymbol{r}'(t)|\, dt. \tag{1.2}$$

If $\varphi : [\tilde{\alpha}, \tilde{\beta}] \to [\alpha, \beta]$ sending $\tilde{t}$ to $t = \varphi(\tilde{t})$ is a coordinate transformation (which is a C^1 map with C^1 inverse, not necessarily increasing), then

$$\frac{d\tilde{\boldsymbol{r}}}{d\tilde{t}} = \frac{d\boldsymbol{r}}{dt} \cdot \frac{dt}{d\tilde{t}},$$

therefore

$$\int_{\varphi^{-1}(\alpha)}^{\varphi^{-1}(\beta)} f\left(\boldsymbol{r}\left(\varphi(\tilde{t})\right)\right) \frac{\left|\tilde{\boldsymbol{r}}'(\tilde{t})\right|}{\left|\varphi'(\tilde{t})\right|} \varphi'(\tilde{t}) d\tilde{t} = \int_{\tilde{\alpha}}^{\tilde{\beta}} f\left(\tilde{\boldsymbol{r}}(\tilde{t})\right) \left|\tilde{\boldsymbol{r}}'(\tilde{t})\right| d\tilde{t}$$

no matter φ is increasing or decreasing.

In the special case when L is given by a graph $y = y(x)$ ($x \in [a, b]$), then

$$\int_L f(x, y) ds = \int_a^b f\left(x, y(x)\right) \sqrt{1 + y'(x)^2} dx.$$

In the special case L is represented as $r = r(\theta)$ ($\alpha \le \theta \le \beta$), then $\boldsymbol{r}(\theta) = (r(\theta)\cos\theta, r(\theta)\sin\theta)$ is a parameterization of the curve, and

$$\left|\boldsymbol{r}'(\theta)\right|^2 = \left(r'(\theta)\cos\theta - r(\theta)\sin\theta\right)^2 + \left(r'(\theta)\sin\theta + r(\theta)\cos\theta\right)^2 = r^2(\theta) + r'^2(\theta)$$

This is equivalent to say that the line element of $\mathbf{R}^2$ is given by

$$ds^2 = dr^2 + r^2 d\theta^2$$

in the polar coordinate. Therefore,

$$\int_L f(x, y) ds = \int_\alpha^\beta f\left(r(\theta)\cos\theta, r(\theta)\sin\theta\right) \sqrt{r^2(\theta) + r'^2(\theta)} d\theta.$$

Example 1.1. Let L be the arc of the eclipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ in the first quadrant. Calculate $\int_L xyds$.

Solution. *Let $(x, y) = (a\cos\theta, b\sin\theta)$ with $\theta \in [0, \frac{\pi}{2}]$ be a parameterization of L, then we get*

$$\begin{aligned}\int_L xyds &= \int_0^{\frac{\pi}{2}} ab\cos\theta\sin\theta\sqrt{a^2\sin^2\theta + b^2\cos^2\theta} d\theta \\ &= \frac{ab}{3(a^2 - b^2)} \left[b^2 + (a^2 - b^2)\sin^2\theta\right]^{\frac{3}{2}} \Big|_0^{\frac{\pi}{2}} \\ &= \frac{ab(a^2 + ab + b^2)}{3(a + b)}.\end{aligned}$$

Example 1.2. Let L be the triangle connecting $O(0, 0)$, $A(1, 0)$ and $B(0, 1)$.Find the line integral $\int_L xds$.

Solution. *This is a piecewisely smooth curve, so we use different parameterization for different pieces.*

We have the following parameterizations

$$\begin{aligned}\overline{OA}: x = t, y = 0, ds = dt, t \in [0,1]\\ \overline{AB}: x = 1-t, y = t, ds = \sqrt{2}dt, t \in [0,1]\\ \overline{BO}: x = 0, y = 1-t, ds = dt, t \in [0,1].\end{aligned}$$

Thus,

$$\begin{aligned}\int_L xds &= \int_{\overline{OA}} xds + \int_{\overline{AB}} xds + \int_{\overline{BO}} xds\\ &= \int_0^1 tdt + \int_0^1 \sqrt{2}(1-t)dt\\ &= \frac{1+\sqrt{2}}{2}.\end{aligned}$$

1.2 Line integral of vector fields

Consider the physical situation that a varying force $\boldsymbol{F}$ is acting on a point mass along a space curve γ, what is the total "work" done by $\boldsymbol{F}$ along this movement?

Since the actual movement is not only the trace of this curve, but also the time parameterization $\boldsymbol{r} : [\alpha, \beta] \longrightarrow \mathbf{R}^3$ encoded in the real physical situation, the curve has a natural **orientation** induced from "timing", i.e. which direction is forwards and which direction is backwards at each point of the path.

To calculate the total work, we partition $[\alpha, \beta]$ into small instants $T : \alpha = t_0 < t_1 < \cdots < t_n = \beta$ and sum up all the works done by $\boldsymbol{F}$ on the path $\gamma|_{[t_{i-1},t_i]}$ on which $\boldsymbol{F}$ is almost a constant force:

$$W(T) = \sum_{i=1}^{n} \boldsymbol{F}(\boldsymbol{r}(\xi_i)) \cdot (\boldsymbol{r}(t_i) - \boldsymbol{r}(t_{i-1}))\,.$$

The limit of $W(T)$ as $\|T\| \to 0$ (if exists) is given a notation

$$\int_\gamma \boldsymbol{F} \cdot d\boldsymbol{r},$$

represent the total work done by $\boldsymbol{F}$ along the movement path γ.

Generalizing this physical situation to abstract mathematical setting, we need to first introduce the notion of "orientation" of a curve.

Definition 1.3 (Orientation of a curve)**.** If the curve $L \subset \mathbf{R}^3$ can be covered by finitely many local piece L_α such that L_α admits a regular C^1 parameterization $\boldsymbol{r}^\alpha : I_\alpha \to L_\alpha$ given by $\boldsymbol{r}^\alpha(t^\alpha) = (x^\alpha(t^\alpha), y^\alpha(t^\alpha), z^\alpha(t^\alpha))$ and on each overlapping $\frac{d\boldsymbol{r}^\beta}{dt^\beta}$ is a positive multiple of $\frac{d\boldsymbol{r}^\alpha}{dt^\alpha}$, then we say L is orientable. In this case, the collection of compatible local parameterizations is called an orientation of L.

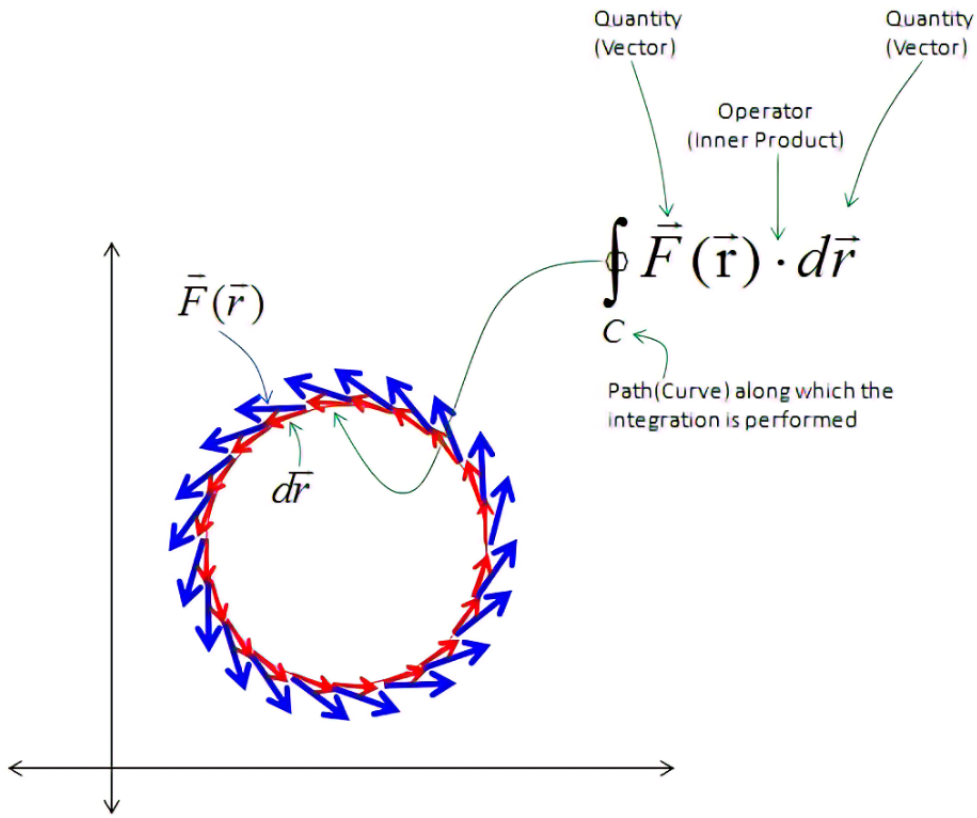


Figure 2: Work done around an oriented closed loop

This is equivalent to the more familiar notion: "L admits a continuous unit tangent vector field".

After defining the orientation of a C^1 curve, we can define the orientation of a piecewisely C^1 curve.

If (L, o) is an oriented curve and $\boldsymbol{v}$ is a vector field on L, then we can form the Riemann sum: take a partition $T : A = M_0, M_1, \cdots, M_n = B$ such that $\overrightarrow{M_{i-1}M_i}$ is in accordance with the orientation o, and take some sampling points $P_i \in L|_{M_{i-1}M_i}$,

$$S(T, \{P_i\}) = \sum_{i=1}^{n} \boldsymbol{v}(P_i) \cdot \overrightarrow{M_{i-1}M_i}.$$

If the limit exists as $\|T\| \to 0$, then we say $\boldsymbol{v}$ is integrable along the curve L, and denote the limit as

$$\int_L \boldsymbol{v} \cdot d\boldsymbol{r},$$

called the line integral of the vector field $\boldsymbol{v}$ along (L, o). In case L is a closed loop, we denote the integral by $\oint_L \boldsymbol{v} \cdot d\boldsymbol{r}$.

It is not difficult to prove that if L is piecewisely C^1 and $\boldsymbol{v}$ is continuous on L, then the above limit exists.

1.3 Computation of line integral of vector field

Suppose the oriented curve (L, o) has a C^1 parameterization $\boldsymbol{r} : [\alpha, \beta] \to \mathbf{R}^3$ such that this parameterization represents the orientation o.

$$\begin{aligned}&\sum_{i=1}^{n} \boldsymbol{v}(\boldsymbol{r}(t_{i-1})) \cdot \Delta \boldsymbol{r}_i \\ &\qquad = \sum_{i=1}^{n} \boldsymbol{v}\left(\boldsymbol{r}(t_{i-1})\right) \cdot \left(\boldsymbol{r}'(t_{i-1}) + o(1)\right) \Delta t_i,\end{aligned}$$

therefore the limit is

$$\int_L \boldsymbol{v} \cdot d\boldsymbol{r} = \int_\alpha^\beta \boldsymbol{v}\left(\boldsymbol{r}(t)\right) \cdot \boldsymbol{r}'(t) dt.$$

1.3.1 Relation with line integral of scalar field

If we choose s to be the particular arc length parametrization (positively oriented) of the curve L, then

$$\int_L \boldsymbol{v} \cdot d\boldsymbol{r} = \int_L \boldsymbol{v} \cdot \boldsymbol{\tau} ds.$$

This is to say that "*the line integral of the vector field $\boldsymbol{v}$ along an oriented curve is the line integral of the scalar field $\boldsymbol{v} \cdot \boldsymbol{\tau}$ along the curve*".

1.3.2 In terms of differential form

In case $\boldsymbol{v} = (P, Q, R) = P(x, y, z)\boldsymbol{i} + Q(x, y, z)\boldsymbol{j} + R(x, y, z)\boldsymbol{k}$, the formula reads as

$$\begin{aligned}\int_L \boldsymbol{v} \cdot d\boldsymbol{r} &= \int_\alpha^\beta \Big(P(x(t), y(t), z(t))x'(t) + Q(x(t), y(t), z(t))y'(t) + R(x(t), y(t), z(t))z'(t)\Big) dt \\ &= \int_L P dx + Q dy + R dz \\ &= \text{“integral of the differential 1-form } P dx + Q dy + R dz \text{ along } L\text{”.}\end{aligned} \tag{1.3}$$

Here dx, dy, dz can be understood as the "oriented projection" of $\boldsymbol{\tau} ds$ into the directions $\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}$ in the following sense. If the direction of the tiny line element is given by

$$\boldsymbol{\tau} = \cos\alpha \boldsymbol{i} + \cos\beta \boldsymbol{j} + \cos\gamma \boldsymbol{k},$$

then

$$\langle dx, \boldsymbol{\tau} ds\rangle = \cos\alpha ds,$$

i.e. when dx acts on a tiny oriented line element $\boldsymbol{\tau} ds$, it measures the "projection" of $\boldsymbol{\tau} ds$ in the x-direction.

1. If $\boldsymbol{v} = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2$, then

   $$\int_L \boldsymbol{v}\cdot d\boldsymbol{r} = c_1\int_L \boldsymbol{v}_1\cdot d\boldsymbol{r} + c_2\int_L \boldsymbol{v}_2\cdot d\boldsymbol{r}.$$

   In particular, take $\boldsymbol{v} = P\boldsymbol{i} + Q\boldsymbol{j} + R\boldsymbol{k} = \boldsymbol{v}_1 + \boldsymbol{v}_2 + \boldsymbol{v}_3$, we obtain that

   $$\int_L \boldsymbol{v}\cdot d\boldsymbol{r} = \int_L Pdx + Qdy + Rdz = \int_L Pdx + \int_L Qdy + \int_L Rdz.$$

   However, $\int_L Pdx$ should NOT be confused with the one-variable integration.

2. If L_{AC} is formed by connecting L_{AB} and L_{BC}, then

   $$\int_{L_{AC}} \boldsymbol{v}\cdot d\boldsymbol{r} = \int_{L_{AB}} \boldsymbol{v}\cdot d\boldsymbol{r} + \int_{L_{BC}} \boldsymbol{v}\cdot d\boldsymbol{r}.$$

   Therefore, if L is piecewisely smooth, then we can calculate the integral piece by piece.

3. Let L_{BA} be the same underlying curve as L_{AB} but with reverse orientation, then

   $$\int_{L_{BA}} \boldsymbol{v}\cdot d\boldsymbol{r} = -\int_{L_{AB}} \boldsymbol{v}\cdot d\boldsymbol{r}.$$

4. If L is $[a, b]$ on the x-axis with positive orientation, then for $\boldsymbol{v} = P\boldsymbol{i} + Q\boldsymbol{j} + R\boldsymbol{k}$ we have

   $$\int_L \boldsymbol{v}\cdot d\boldsymbol{r} = \int_a^b P(x, 0, 0)dx$$

   is the usual one-variable integration.

Example 1.4. Let L be the triangle OAB with anti-clockwise orientation where $A = (1, 0)$, $B = (1, 2)$. Find the line integral $\int_L xydx + x^2dy$.

Example 1.5. Find the work done by the force $\boldsymbol{F} = y\boldsymbol{i} - x\boldsymbol{j} + z\boldsymbol{k}$ along the helix curve $L : x = R\cos t, y = R\sin t, z = kt$ ($t \in [0, 2\pi]$) from $A(R, 0, 0)$ to $B(R, 0, 2\pi k)$.

Example 1.6. The gravity on earth m by the sun M is given by $\boldsymbol{F} = -G\frac{Mm}{r^3}\boldsymbol{r}$ where $\boldsymbol{r}$ is the position vector from M to m. Find the work done by the gravity if m moves from A to B.

Solution. *Let* $\boldsymbol{r} = \boldsymbol{r}(t)$ *be the time-parameter equation for the curve, then*

$$W = \int_{L_{AB}} \boldsymbol{F} \cdot d\boldsymbol{r} = -GMm \int_{L_{AB}} \frac{1}{r^3} \boldsymbol{r} \cdot d\boldsymbol{r}.$$

In the integrand,

$$\boldsymbol{r} \cdot d\boldsymbol{r} = (xx' + yy' + zz')\, dt = \frac{1}{2} \frac{d[r(t)^2]}{dt},$$

therefore,

$$W = -\frac{1}{2} GMm \int_{t_A}^{t_B} \frac{1}{r^3} \frac{d[r(t)^2]}{dt} dt = GMm \left(\frac{1}{r_B} - \frac{1}{r_A} \right).$$

This work depends only on the initial and ending point of the position regardless of which path it takes to achieve such displacement.

Example 1.7. Find the circulation of the two vector fields $\boldsymbol{v}_1 = \frac{-y\boldsymbol{i}+x\boldsymbol{j}}{x^2+y^2}, \boldsymbol{v}_2 = \frac{x\boldsymbol{i}+y\boldsymbol{j}}{x^2+y^2}$ along the anti-clockwise circle $x^2 + y^2 = R^2$.

2 Surface integral

2.1 Surface integral of scalar fields

Suppose there is a very thin surface S in $\mathbf{R}^3$ with surface density given by the function ρ on S, then the total mass of the surface could be represented by

$$\lim_{\|T\|\to 0} \sum_{i=1}^{n} \rho(M_i)\Delta S_i.$$

The limit is denoted by

$$\iint_S \rho dS$$

if it exists.

In general, if $f : S \longrightarrow \mathbf{R}$ is a function, and S is piecewisely C^1 surface, then the integration of f on S is denoted by

$$\iint_S f dS = \iint_S f d\sigma.$$

In case S has a C^1 parametrization $\boldsymbol{r} : D \subset \mathbf{R}^2 \to \mathbf{R}^3$, then the integral is calculated as

$$\iint_D f\left(x(u,v), y(u,v), z(u,v)\right)\sqrt{EG - F^2} dudv, \tag{2.1}$$

where $ds^2 = Edu^2 + 2Fdudv + Gdv^2$ is the first fundamental form of S represented under this parameterization.

If $\varphi : \tilde{D} \to D$ sending $(\tilde{u}, \tilde{v})$ to (u, v) is an invertible coordinate transformation, then $\tilde{\boldsymbol{r}}(\tilde{u}, \tilde{v}) = \boldsymbol{r}(\varphi(\tilde{u}, \tilde{v}))$ gives another parameterization of S. Regarding the coefficients of the first fundamental form, we have

$$\begin{pmatrix} \tilde{E} & \tilde{F} \\ \tilde{F} & \tilde{G} \end{pmatrix} = \begin{pmatrix} \frac{\partial u}{\partial \tilde{u}} & \frac{\partial v}{\partial \tilde{u}} \\ \frac{\partial u}{\partial \tilde{v}} & \frac{\partial v}{\partial \tilde{v}} \end{pmatrix} \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} \frac{\partial u}{\partial \tilde{u}} & \frac{\partial u}{\partial \tilde{v}} \\ \frac{\partial v}{\partial \tilde{u}} & \frac{\partial v}{\partial \tilde{v}} \end{pmatrix},$$

thus $\sqrt{\tilde{E}\tilde{G} - \tilde{F}^2} = \sqrt{EG - F^2}\left|\frac{\partial(u,v)}{\partial(\tilde{u},\tilde{v})}\right|$. The consequence is that

$$\begin{aligned} &\iint_D f(x(u,v), y(u,v), z(u,v))\sqrt{EG - F^2} dudv \\ &\qquad = \iint_{\tilde{D}} f(x(\tilde{u},\tilde{v}), y(\tilde{u},\tilde{v}), z(\tilde{u},\tilde{v}))\sqrt{\tilde{E}\tilde{G} - \tilde{F}^2} d\tilde{u}d\tilde{v} \end{aligned}$$

by the substitution formula. In other words, the surface integral *does not depend on the particular choice of a parametrization*.

2.2 Surface integral of vector fields

In physical background the notion of "flux" through a surface is very important: i.e. for instance the volume of water pssing through a surface in unit time, the magnetism field passing through a surface in unit time..... There are two key ingredients in the determination of the "flux":

- Which side is "positive side" and which is "negative side" for the surface?
- The orthogonal projection of the field $\boldsymbol{v}$ to the "positive normal direction" $\boldsymbol{n}$.

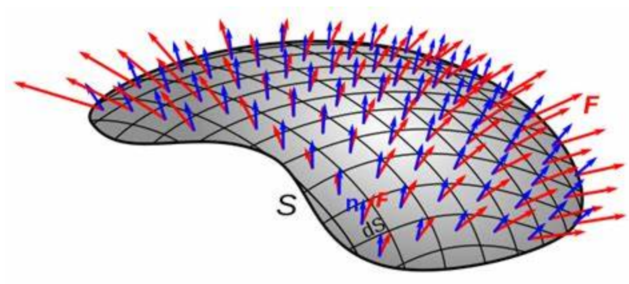


Figure 3: Flux through a surface

Regarding the first ingredient, we need to define the "orientation" of a surface:

Definition 2.1 (Orientation of a surface). If $S \subset \mathbf{R}^3$ can be covered by finitely many open pieces S_α such that each piece admits a regular C^1 parameterization $\boldsymbol{r}^\alpha : U_\alpha \to S_\alpha$ given by $\boldsymbol{r}^\alpha(u_\alpha, v_\alpha) = (x^\alpha(u^\alpha, v^\alpha), y^\alpha(u^\alpha, v^\alpha), z^\alpha(u^\alpha, v^\alpha))$ and on each overlap $S_\alpha \cap S_\beta$, $\frac{\partial(u^\alpha, v^\alpha)}{\partial(u^\beta, v^\beta)} > 0$($\boldsymbol{r}^\alpha$ and $\boldsymbol{r}^\beta$ are called compatible if this happens), then we say S is orientable. In this case, the collection of compatible local parameterizations is called the induced orientation.

Since in $\mathbf{R}^3$, the normal vector at a point is represented by the *cross product* and $\frac{\partial \boldsymbol{r}^\beta}{\partial u^\beta} \times \frac{\partial \boldsymbol{r}^\beta}{\partial v^\beta} = \frac{\partial(u^\alpha, v^\alpha)}{\partial(u^\beta, v^\beta)} \frac{\partial \boldsymbol{r}^\alpha}{\partial u^\alpha} \times \frac{\partial \boldsymbol{r}^\alpha}{\partial v^\alpha}$, the above definition of "orientable" is equivalent to the more extrinsic notion: "S admits a continuous unit normal vector field $\boldsymbol{n}$".

If $(S, \boldsymbol{n})$ is an oriented surface, then $(S, -\boldsymbol{n})$ is also an oriented surface with the reversed orientation. In $\mathbf{R}^3$, a connected orientable surface has two orientations.

One should remark that "orientability" is an intrinsic property of two dimensional surface since compatibility can be represented by all Jacobians $\frac{\partial(u^\alpha, v^\alpha)}{\partial(u^\beta, v^\beta)} > 0$. The "two-sided property" of a two dimensional surface (either locally or globally) is a property adhering to "codimension one" setting. In higher codimension situation, this notion does not make much sense as the example of curves in $\mathbf{R}^3$ shows.

Example 2.2 (Hypersurface in $\mathbf{R}^3$ defined by one equation). Let $S = \{(x, y, z) \in \mathbf{R}^3 | F(x, y, z) = 0\}$ be a level set of the C^1 function F on $\mathbf{R}^3$. Supoose there is no critical point of F on S, then S is an orientable surface with one orientation given by the continuous unit normal vector field $\frac{\nabla F}{|\nabla F|}$.

Example 2.3 (Non-orientable surface). Consider two vector-valued maps

$$\boldsymbol{r}_1 : (\theta, r) \mapsto (\cos\theta, \sin\theta, 0) + r\left[\sin\frac{\theta}{2}\cdot(-\cos\theta, \sin\theta, 0) + \cos\frac{\theta}{2}\cdot(0, 0, 1)\right],$$
$$\theta \in (0, 2\pi), r \in (-1, 1);$$
$$\boldsymbol{r}_2 : (\varphi, s) \mapsto (\cos\varphi, \sin\varphi, 0) + s\left[\sin\frac{\varphi}{2}\cdot(-\cos\varphi, \sin\varphi, 0) + \cos\frac{\varphi}{2}\cdot(0, 0, 1)\right],$$
$$\varphi \in (-\pi, \pi), s \in (-1, 1).$$

The image of the two maps glues together giving a Mobius band. The transition map is

$$(\varphi, s) \mapsto (\theta, r) = (\varphi + 2\pi, -s), \;\; \varphi \in (-\pi, 0), s \in (-1, 1)$$
$$(\varphi, s) \mapsto (\theta, r) = (\varphi, s), \;\; \varphi \in (0, \pi), s \in (-1, 1).$$

Therefore, the two parameterizations are not compatible. Actually, Mobius band is non-orientable, meaning that we cannot find any continuous unit normal vector field (of course defined everywhere) on it.

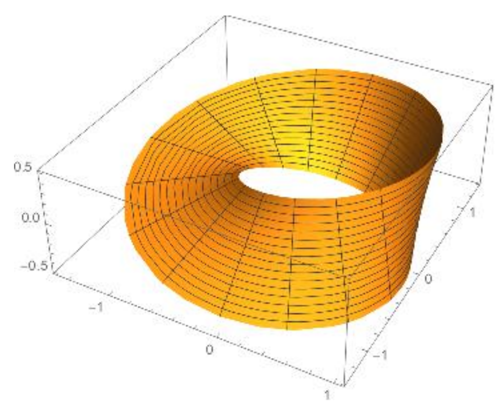


Figure 4: Mobius band with the given parameterization

With orientation given on the surface S, the quantity/volume passing through a tiny piece ΔS in a tiny time period Δt can be represented as

$$\Delta V = \boldsymbol{v}\Delta t \cdot \boldsymbol{n}\Delta S + o(\Delta S\Delta t).$$

Using the Riemann sum, we can get the total flux:

$$\Phi = \iint_S \boldsymbol{v}\cdot\boldsymbol{n}dS = \iint_S \boldsymbol{v}\cdot d\boldsymbol{S}. \tag{2.2}$$

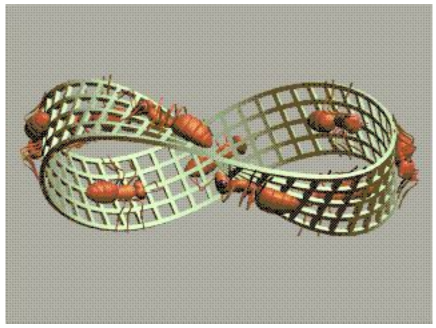


Figure 5: Living on a Mobius band: which side is upper and which is down?

In this expression, $d\boldsymbol{S} = \boldsymbol{n} dS$ is called an "oriented area element". If the surface is closed, i.e. without boundary, we also denote it by

$$\oint_S \boldsymbol{v} \cdot d\boldsymbol{S}.$$

If $\boldsymbol{r}(u,v) = (x(u,v), y(u,v), z(u,v))$ where $(u,v) \in D \subset \mathbf{R}^2$ is consistent parameterization, i.e. $\boldsymbol{r}'_u \times \boldsymbol{r}'_v$ is a positive multiple of $\boldsymbol{n}$ everywhere, then we have

$$\boldsymbol{n} = \frac{\boldsymbol{r}'_u \times \boldsymbol{r}'_v}{|\boldsymbol{r}'_u \times \boldsymbol{r}'_v|}, \quad dS = |\boldsymbol{r}'_u \times \boldsymbol{r}'_v|\, dudv,$$

so

$$\begin{aligned} \iint_S \boldsymbol{v} \cdot d\boldsymbol{S} &= \iint_D \boldsymbol{v} \cdot (\boldsymbol{r}'_u \times \boldsymbol{r}'_v)\, dudv \\ &= \iint_D \left(P \frac{\partial(y,z)}{\partial(u,v)} + Q \frac{\partial(z,x)}{\partial(u,v)} + R \frac{\partial(x,y)}{\partial(u,v)} \right) dudv. \end{aligned} \tag{2.3}$$

2.3 differential form interpretation

Let P, Q, R be functions on $\mathbf{R}^3$, then the expression $\omega = Pdy \wedge dz + Qdz \wedge dx + Rdx \wedge dy$ is called a differential 2-form. At each point $\boldsymbol{x} \in \mathbf{R}^3$, $\omega|_{\boldsymbol{x}}$ is an **antisymmetric bilinear form** on $T_{\boldsymbol{x}}\mathbf{R}^3$, whose matrix under the standard basis $\{\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}\}$ of $T_{\boldsymbol{x}}\mathbf{R}^3$ is given as

$$\begin{pmatrix} 0 & R & -Q \\ -R & 0 & P \\ Q & -P & 0 \end{pmatrix}$$

Heuristically, it acts on the "oriented area element" $\Delta\boldsymbol{S} = \boldsymbol{r}'_u \Delta u \times \boldsymbol{r}'_v \Delta v$ according to anti-symmetric property and linearity, i.e

$$
\begin{aligned}
\omega(\Delta \boldsymbol{S}) &= \omega(\boldsymbol{r}'_u \Delta u, \boldsymbol{r}'_v \Delta v) \\
&= P\left(dy(\boldsymbol{r}'_u \Delta u) dz(\boldsymbol{r}'_v \Delta v) - dz(\boldsymbol{r}'_u \Delta u) dy(\boldsymbol{r}'_v d\Delta v)\right) \\
&\quad + Q\left(dz(\boldsymbol{r}'_u \Delta u) dx(\boldsymbol{r}'_v \Delta v) - dx(\boldsymbol{r}'_u \Delta u) dz(\boldsymbol{r}'_v \Delta v)\right) \\
&\qquad + R\left(dx(\boldsymbol{r}'_u \Delta u) dy(\boldsymbol{r}'_v \Delta v) - dy(\boldsymbol{r}'_u \Delta u) dx(\boldsymbol{r}'_v \Delta v)\right) \\
&= \left(P(y'_u z'_v - z'_u y'_v) + Q(z'_u x'_v - x'_u z'_v) + R(x'_u y'_v - y'_u z'_v)\right) \Delta u \Delta v,
\end{aligned} \tag{2.4}
$$

during which we use our previous understanding about differential of the coordinate functions x, y, z, i.e. that $dx(\boldsymbol{r}'_n) = dx(x'_u \boldsymbol{i} + y'_u \boldsymbol{j} + z'_u \boldsymbol{k}) = x'_u$ etc.

How to under that "$dy \wedge dz$ is the projection of $d\boldsymbol{S}$ to yOz plane, etc?"

Let Π be a plane defined by $\boldsymbol{n} \cdot \boldsymbol{r} = d$ in $\mathbf{R}^3$, and S be a region on this plane with orientation given by $\boldsymbol{n}$. Denote S_{xy} to be the projection of S onto xOy plane. Suppose $n_3 > 0$, then $(x, y) \in S_{xy} \to (x, y, \frac{d - n_1 x - n_2 y}{n_3})$ is a positive parametrization of S. We have

$$
\text{Area}(S) = \iint_{S_{xy}} \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}\, dx dy = \iint_{S_{xy}} \frac{1}{|n_3|} dx dy = \frac{\text{Area}(S_{xy})}{|n_3|}.
$$

Therefore, $\text{Area}(S_{xy}) = n_3 \cdot \text{Area}(S)$. And a consequence is that

$$
\text{Area}(S_{yz})^2 + \text{Area}(S_{zx})^2 + \text{Area}(S_{xy})^2 = \text{Area}(S)^2, \tag{2.5}
$$

which can be viewed as Pythagoras' Theorem for two dimensional object.

Combining the computation formula (2.3) and the rule (2.4), we get that

$$
\begin{aligned}
\iint_S \boldsymbol{v} \cdot d\boldsymbol{S} &= \iint_S P dy \wedge dz + Q dz \wedge dx + R dx \wedge dy \\
&= \text{“integral of the differential 2-form } P dy \wedge dz + Q dz \wedge dx + R dx \wedge dy \text{ along } S\text{”}.
\end{aligned} \tag{2.6}
$$

On the other hand, using the formula

$$
\begin{aligned}
dx &= x'_u du + x'_v dv \\
dy &= y'_u du + y'_v dv \\
dz &= z'_u du + z'_v dv
\end{aligned}
$$

and the property that $du \wedge du = dv \wedge dv = 0$ and $dv \wedge du = -du \wedge dv$, we get

$$
\begin{aligned}
P dy \wedge dz + Q dz \wedge dx + R dx \wedge dy &= P\left(y'_u du + y'_v dv\right) \wedge \left(z'_u du + z'_v dv\right) \\
&\quad + Q\left(z'_u du + z'_v dv\right) \wedge \left(x'_u du + x'_v dv\right) \\
&\qquad + R\left(x'_u du + x'_v dv\right) \wedge \left(y'_u du + y'_v dv\right) \\
&= \left(P \frac{\partial(y, z)}{\partial(u, v)} + Q \frac{\partial(z, x)}{\partial(u, v)} + R \frac{\partial(x, y)}{\partial(u, v)}\right) du \wedge dv.
\end{aligned} \tag{2.7}
$$

This means that for computational convenience we can simply plug the parameterization directly into the 2-form and simplify the integrand into the usual notion of double integral according to the rule of wedge product "$\wedge$".

A special case is when S is actually a surface included in the xOy plane, and we take the $\boldsymbol{k}$ as its orientation, then

$$\iint_S Pdy\wedge dz+Qdz\wedge dx+Rdx\wedge dy=\iint_D R(x,y,0)dx\wedge dy=\iint_D R(x,y,0)dxdy$$

where the last expression is the usual double integral. Therefore, the usual double integral could be understood as a surface integral (where the orientation of the planar region D is taken to be the natural anticlockwise orientation):

$$\iint_D f(u,v)dudv=\iint_D f(u,v)du\wedge dv.$$

Example 2.4. Suppose S is an oriented surface (closed or not) with unit normal vector field given by $\boldsymbol{n}=(n_1,n_2,n_3)$, then

$$\iint_S n_1dy\wedge dz+n_2dz\wedge dx+n_3dx\wedge dy=\iint_S \boldsymbol{n}\cdot d\boldsymbol{S}=\iint_S dS,$$

where the first two integration domain S is assumed to be have the given orientated, and the last integral is integral of scalar function. The importance of this formula is reflected by the fact that the left hand side provides a formula for the area of S in terms of implicit characterization of S instead of using explicit parameterization.

More generally, for any f defined on S,

$$\iint_S fdS=\iint_S f\boldsymbol{n}\cdot d\boldsymbol{S},$$

i.e. the scalar integration could be regarded as a special case of vector integral on surface.

Example 2.5. Find the flux of the vector field $\boldsymbol{v}(x,y,z)=P(x,y,z)\boldsymbol{i}+Q(x,y,z)\boldsymbol{j}+R(x,y,z)\boldsymbol{k}$ from inner of the sphere $S:x^2+y^2+z^2=a^2$ to outside.

Solution. *The unit normal vector representing the orientation should be* $\boldsymbol{n}=\frac{1}{a}(x,y,z)$, *thus the total flux is*

$$\oint_S \boldsymbol{v}\cdot d\boldsymbol{S}=\oint_S \boldsymbol{v}\cdot\boldsymbol{n}dS=\frac{1}{a}\iint_S(xP+yQ+zR)\,dS.$$

For instance, for electric field $\boldsymbol{E}=\frac{q}{r^3}\boldsymbol{r}$ *generated by a point charge* q *located at* $\mathbf{0}$, *the total flux is*

$$\oint_S \boldsymbol{E}\cdot d\boldsymbol{S}=\int_S\frac{q}{r^3}\boldsymbol{r}\cdot\frac{r}{a}dS=\int_S\frac{qr^2}{ar^3}dS=4\pi q.$$

One important observation is that this flux is independent of radius of the sphere.

Example 2.6. Let S be the cylinder $x^2 + y^2 = a^2, -h \leq z \leq h$ with outward orientation, represent the flux for the vector field $\boldsymbol{v}(x,y,z) = P(x,y,z)\boldsymbol{i} + Q(x,y,z)\boldsymbol{j} + R(x,y,z)\boldsymbol{k}$.

Solution. *We can take $\boldsymbol{n} = \frac{1}{a}(x\boldsymbol{i} + y\boldsymbol{j})$ and parametrize the surface by cylindrical coordinate $x = a\cos\theta, y = a\sin\theta, z = z$ $(0 \leq \theta \leq 2\pi,\ -h \leq z \leq h)$. Under this parameterization, $\boldsymbol{r}_\theta = (-a\sin\theta, a\cos\theta, 0)$ and $\boldsymbol{r}_z = (0,0,1)$ and therefore $dS = |\boldsymbol{r}_\theta \times \boldsymbol{r}_z| d\theta dz = a d\theta dz$.*

The total flux can be represented as

$$\int_{0\leq\theta\leq 2\pi, -h\leq z\leq h} (P\cos\theta + Q\sin\theta) a d\theta dz.$$

As a particular example,

$$\begin{aligned}\iint_S \boldsymbol{E}\cdot d\boldsymbol{S} &= \int_{0\leq\theta\leq 2\pi, -h\leq z\leq h} (P\cos\theta + Q\sin\theta) a d\theta dz \\ &= \int_{0\leq\theta\leq 2\pi, -h\leq z\leq h} \frac{a^2 q}{(a^2+z^2)^{\frac{3}{2}}} d\theta dz \\ &= 4\pi q \frac{h}{\sqrt{a^2+h^2}}.\end{aligned}$$

Try to understand what is the physical meaning of this result as the parameters a, h vary? 立体角的解释。

Example 2.7. Let $\boldsymbol{v} = y(x-z)\boldsymbol{i} + x^2\boldsymbol{j} + (y^2+xz)\boldsymbol{k}$, and S be the boundary of the cuboid: $0 \leq x \leq a, 0 \leq y \leq b, 0 \leq z \leq c$ oriented by the outward normal. Find the flux of $\boldsymbol{v}$ through S.

Solution. *The surface is piecewisely smooth, so we can decompose the whole surface integral into six pieces.*

$$\begin{aligned}\int_S \boldsymbol{v}\cdot d\boldsymbol{S} &= \int_{S_1} y(x-z) dy\wedge dz + \int_{S_2} y(x-z) dy\wedge dz \\ &\quad + \int_{S_3} x^2 dz\wedge dx + \int_{S_4} x^2 dz \wedge dx \\ &\qquad + \int_{S_5} (y^2+xz) dx\wedge dy + \int_{S_6} (y^2+xz)\wedge dy\wedge dz \\ &= \iint_{0\leq y\leq b, 0\leq z\leq c} \big(y(a-z) - y(0-z)\big) dydz \\ &\quad + \iint_{0\leq z\leq c, 0\leq x\leq a} \big(x^2 - x^2\big) dzdx \\ &\qquad + \iint_{0\leq x\leq a, 0\leq y\leq b} \big((y^2+cx) - (y^2 + 0\cdot x)\big) dxdy \\ &= \frac{1}{2} abc(a+b).\end{aligned}$$

Example 2.8. Find the surface integral $\iint_S x^2dydz + y^2dzdx + z^2dxdy$ where S is the hemisphere of $x^2 + y^2 + z^2 = a^2$ $(z \geq 0)$ with upward orientation.

Solution. *We can simply take the spherical coordinate $x = a\sin\theta\cos\varphi, y = a\sin\theta\sin\varphi, z = a\cos\theta$ $(\theta \in [0, \frac{\pi}{2}], \varphi \in [0, 2\pi])$. We can check that*

$$\boldsymbol{r}_\theta \times \boldsymbol{r}_\varphi = \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ a\cos\theta\cos\varphi & a\cos\theta\sin\varphi & -a\sin\theta \\ -a\sin\theta\sin\varphi & a\sin\theta\cos\varphi & 0 \end{vmatrix} = a^2\sin\theta\left(\sin\theta\cos\varphi\boldsymbol{i} + \sin\theta\sin\varphi\boldsymbol{j} + \cos\theta\boldsymbol{k}\right)$$

points outwards which is upwards in the north hemisphere S, therefore (θ, φ) is a positive parametrization. The surface integral is thus

$$\begin{aligned}
&\iint_S \left(x^2\boldsymbol{i} + y^2\boldsymbol{j} + z^2\boldsymbol{k}\right) \cdot \boldsymbol{n}dS \\
&\quad= \iint \left(a^2\sin^2\theta\cos^2\varphi \cdot \sin\theta\cos\varphi + a^2\sin^2\theta\sin^2\varphi \cdot \sin\theta\sin\varphi + a^2\cos^2\theta \cdot \cos\theta\right) a^2\sin\theta d\theta d\varphi \\
&\quad= \int_0^{\frac{\pi}{2}} a^4\cos^3\theta\sin\theta d\theta \int_0^{2\pi} d\varphi \\
&\quad= \frac{1}{2}\pi a^4.
\end{aligned}$$

The textbook takes another way of observing the symmetry of the surface S with respected to the yOz plane and zOx plane. The function of integration is even why the Euclidean coordinate functions provides positive orientation on one half of the surface and negative orientation on its mirror half. Therefore, only $z^2dx \wedge dy$ survives, which we could calculate directly $\iint_{x^2+y^2 \leq a^2}(a^2 - x^2 - y^2)dxdy$.

Example 2.9. Find the surface integral $\iint_S x^3dydz + y^3dzdx$.

3 Green's Theorem, Stokes' Theorem and Gauss Theorem

3.1 Green's Theorem

The circulation of a planar vector field along a planar closed curve can be transformed to certain double integral.

Let $\boldsymbol{v} = P(x,y)\boldsymbol{i} + Q(x,y)\boldsymbol{j}$ be a continuous vector field defined on a planar region $D \subset \mathbf{R}^2$. Let us first consider several regions of simple shape.

1. **[Case 1: rectangular region]** Let $D = [a,b] \times [c,d]$ equipped with anticlockwise orientation. If furthermore $\frac{\partial P}{\partial y}, \frac{\partial Q}{\partial x}$ are continuous on D, then

$$\begin{aligned}
\oint_{\partial D} Pdx + Qdy &= \left(\int_{L_1} + \int_{L_2} + \int_{L_3} + \int_{L_4}\right) Pdx + Qdy \\
&= \int_a^b P(x,c)dx + \int_c^d Q(b,y)dy - \int_a^b P(x,d)dx - \int_c^d Q(a,y)dy \\
&= \int_c^d (Q(b,y) - Q(a,y))\, dy - \int_a^b (P(x,d) - P(x,c))\, dx \\
&= \int_c^d dy \int_a^b \frac{\partial Q}{\partial x}(x,y)dx - \int_a^b dx \int_c^d \frac{\partial P}{\partial y}(x,y)dy \\
&= \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dxdy.
\end{aligned}$$

2. **[Case 2: Type A region]** Suppose $D = \{(x,y) | a \le x \le b, c \le y \le \psi(x)\}$ where ψ is monotone and C^1, with differentiable inverse. Without loss of generality, assume ψ is strictly increasing, then

$$\begin{aligned}
\oint_{\partial D} Pdx + Qdy &= \int_a^b P(x,c)dx + \int_c^{\psi(b)} Q(b,y)dy \\
&\quad - \int_a^b (P(x,\psi(x)) + Q(x,\psi(x))\psi'(x))\, dx - \int_c^{\psi(a)} Q(a,y)dy \\
&= -\int_a^b dx \int_c^{\psi(x)} \frac{\partial P}{\partial y}(x,y)dy + \int_c^{\psi(a)} dy \int_a^b \frac{\partial Q}{\partial x}(x,y)dx \\
&\quad + \int_{\psi(a)}^{\psi(b)} Q(b,y)dy - \int_a^b Q(x,\psi(x))\psi'(x)dx \\
&= -\iint_D \frac{\partial P}{\partial y}(x,y)dxdy + \int_a^b dx \int_c^{\psi(a)} \frac{\partial Q}{\partial x} dy + \int_{\psi(a)}^{\psi(b)} dy \int_{\psi^{-1}(y)}^b \frac{\partial Q}{\partial x} dx \\
&= \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dxdy.
\end{aligned}$$

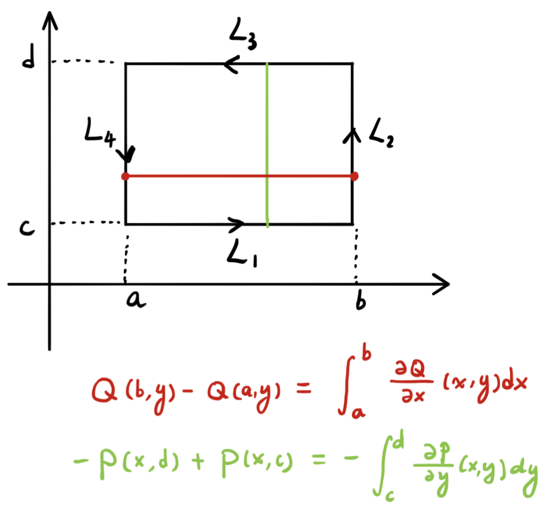
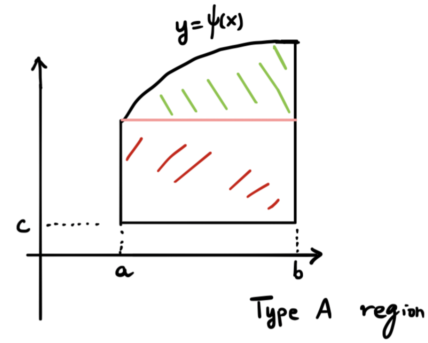


Figure 6: Rectangular region

3. **[Case 3: Type B region]** Suppose $D = \{(x,y)|a \le x \le b, \psi(x) \le y \le d\}$ where ψ is monotone and C^1, with differentiable inverse. Without loss of generality, assume ψ is strictly decreasing, then

$$\begin{aligned}
\oint_{\partial D} Pdx + Qdy &= \int_a^b \big(P(x,\psi(x)) + Q(x,\psi(x))\psi'(x)\big)dx \\
&\quad + \int_{\psi(b)}^d Q(b,y)dy - \int_a^b P(x,d)dx - \int_{\psi(a)}^d Q(a,y)dy \\
&= -\int_a^b dx \int_{\psi(x)}^d \frac{\partial P}{\partial y} dy + \int_{\psi(a)}^d dy \int_a^b \frac{\partial Q}{\partial x} dx \\
&\quad + \int_{\psi(b)}^{\psi(a)} Q(b,y)dy + \int_a^b Q(x,\psi(x))\psi'(x)dx \\
&= -\iint_D \frac{\partial P}{\partial y} dxdy + \int_a^b dx \int_{\psi(a)}^d \frac{\partial Q}{\partial x} dy + \int_{\psi(b)}^{\psi(a)} dy \int_{\psi^{-1}(y)}^b \frac{\partial Q}{\partial x} dx \\
&= \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dxdy
\end{aligned}$$

4. For regions of the shape $\{(x,y) \in \mathbf{R}^2 | c \le y \le d, a \le x \le \varphi(y)\}$ and $\{(x,y) \in \mathbf{R}^2 | c \le y \le d, \phi(y) \le x \le b\}$ with monotone function φ, we can prove the above formula similarly.



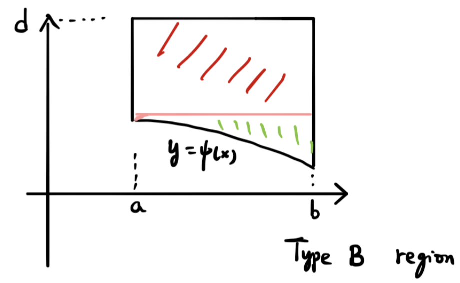


Figure 7: Type A and Type B region

Theorem 3.1 (Green's Theorem)**.** *Suppose $D \subset \mathbf{R}^2$ can be divided into the union of finitely many regions D_i's of the types considered above, such that D_i and D_j has no common interior. Let $\boldsymbol{v} = P(x,y)\boldsymbol{i} + Q(x,y)\boldsymbol{j}$ be a continuous vector field on D such that $\frac{\partial P}{\partial y}$ and $\frac{\partial Q}{\partial x}$ are continuous on D, then*

$$\oint_{\partial D} \boldsymbol{v} \cdot d\boldsymbol{r} = \oint_{\partial D} Pdx + Qdy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dxdy,$$

where the curve $\partial D = \cup_{k=1}^{N} L_k$ takes the orientation such that D is on the left hand side if you are heading to the orientation.

We only need to notice that all the boundary pieces of D_i's appear twice if it lies inside D and the two times appears in opposite orientations. Applying Green's Theorem to this subdomains and sum them together, we arrive at the conclusion of Green's Theorem.

Remark 3.2.

- The common region we encounter satisfies the condition of Green's Theorem.

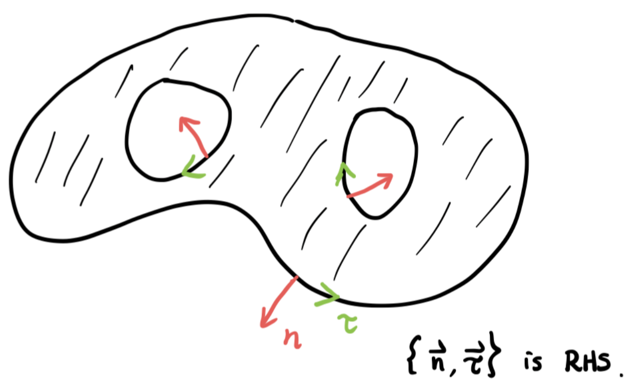


Figure 8: Boundary orientation in Green's Theorem

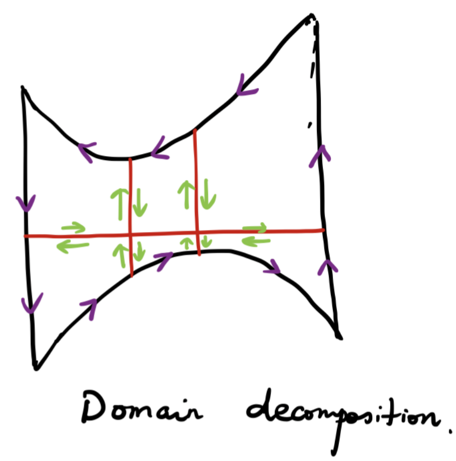


Figure 9: Decomposition of the region for the proof of Green's Theorem

- More generally, if ∂D consists of finitely many piecewisely C^1 curves, then Green's Theorem holds. The proof can be divided into the following steps:
  - *Prove the Green's Theorem for polygonal boundary;*
  - *Prove that* $\boldsymbol{v}$ *could be extended to a neighborhood of* $\overline{D}$ *such that the extended vector field still satisfies the regularity condition, i.e.* $\boldsymbol{v}$ *and* $\frac{\partial P}{\partial y}, \frac{\partial Q}{\partial x}$ *are continuous in an open neighborhood of* $\overline{D}$*;*

– *Approximate ∂D by a sequence of polygonal curves γ_k which encloses regions D_k. Prove that*

$$\oint_{\gamma_k} Pdx + Qdy \to \oint_{\partial D} Pdx + Qdy$$

and

$$\iint_{D_k} \left(\frac{\partial Q}{\partial y} - \frac{\partial P}{\partial x}\right) dxdy \to \iint_{D} \left(\frac{\partial Q}{\partial y} - \frac{\partial P}{\partial x}\right) dxdy$$

as $k \to +\infty$.

Example 3.3. Let D be a region satisfying the condition of Green's Theorem, and $\boldsymbol{v}_1 = x\boldsymbol{j}, \boldsymbol{v}_2 = -y\boldsymbol{i}$ and $\boldsymbol{v}_3 = \frac{1}{2}\left(-y\boldsymbol{i} + x\boldsymbol{j}\right)$ be three smooth vector fields on $\mathbf{R}^2$. We then have

$$\begin{aligned}
\oint_{\partial D} \boldsymbol{v}_1 \cdot d\boldsymbol{r} &= \oint_{\partial D} xdy = \iint_D \frac{\partial}{\partial x}(x)dxdy = \iint_D dxdy = \text{Area}(D) \\
\oint_{\partial D} \boldsymbol{v}_2 \cdot d\boldsymbol{r} &= \oint_{\partial D} -ydx = \iint_D -\frac{\partial}{\partial y}(-y)dxdy = \iint_D dxdy = \text{Area}(D) \\
\oint_{\partial D} \boldsymbol{v}_3 \cdot d\boldsymbol{r} &= \frac{1}{2}\oint_{\partial D} -ydx + xdy = \frac{1}{2}\int_{\partial D} (\boldsymbol{v}_1 + \boldsymbol{v}_2) \cdot d\boldsymbol{r} = \text{Area}(D).
\end{aligned}$$

Let $D : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ be an eclipse, then

$$\begin{aligned}
\text{Area}(D) &= \frac{1}{2}\oint_{\partial D} -ydx + xdy = \frac{1}{2}\int_0^{2\pi} \big(-b\sin\theta \cdot (-a\sin\theta) + a\cos\theta \cdot (b\cos\theta)\big)d\theta \\
&= \pi ab.
\end{aligned}$$

If L is a simple planar polygon formed by successively (anticlockwise) connecting $M_i = (x_i, y_i)$, $i = 1, 2, \cdots, n$. Find the area of the region enclosed by L.

$$\begin{aligned}
\frac{1}{2}\oint_L xdy - ydx &= \frac{1}{2}\sum_{i=1}^{n}\int_{L_i} xdy - ydx \\
&= \frac{1}{2}\sum_{i=1}^{n}\int_0^1 \Big([(1-t)x_i + tx_{i+1}](y_{i+1} - y_i) - [(1-t)y_i + ty_{i+1}](x_{i+1} - x_i)\Big)dt \\
&= \frac{1}{2}\sum_{i=1}^{n}\begin{vmatrix} x_i & y_i \\ x_{i+1} & y_{i+1} \end{vmatrix}
\end{aligned}$$

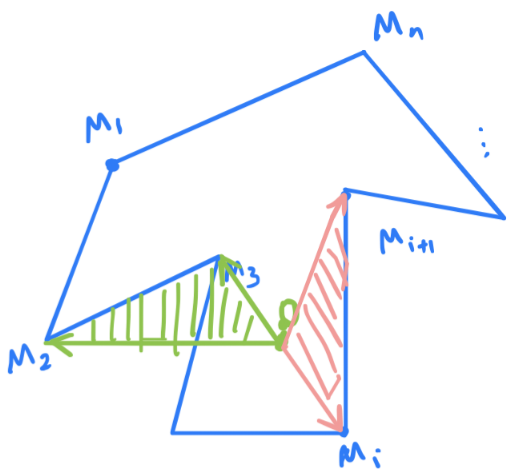


Figure 10: Polygonal region

Example 3.4. Let L be a simple closed C^1 curve around $(0,0)$ with anticlockwise orientation, compute the line integral

$$\oint_L \frac{-ydx + xdy}{x^2+y^2}.$$

Solution. *The vector field* $\boldsymbol{v} = -\frac{y}{x^2+y^2}\boldsymbol{i} + \frac{x}{x^2+y^2}\boldsymbol{j}$ *has a singularity at* $(0,0)$, *therefore not continuous in the region enclosed by* L. *We cannot apply Green's Theorem directly to the region enclosed by* L.

If we take D_ε *to be the region enclosed by* L *and the* $\partial B_\varepsilon(\mathbf{0})$, *then the vector field is* C^1 *in* D_ε. *Apply Green's Theorem, we obtain that*

$$\begin{aligned}
\oint_{\partial D_\varepsilon} \frac{-ydx + xdy}{x^2+y^2} &= \oint_L \frac{-ydx + xdy}{x^2+y^2} - \oint_{\partial B_\varepsilon(\mathbf{0})} \frac{-ydx + xdy}{x^2+y^2} \\
&= \iint_{D_\varepsilon} \left(\frac{\partial}{\partial x}\frac{x}{x^2+y^2} + \frac{\partial}{\partial y}\frac{y}{x^2+y^2} \right) dxdy \\
&= \iint_{D_\varepsilon} \frac{(x^2+y^2) - 2x^2 + (x^2+y^2) - 2y^2}{(x^2+y^2)^2} dxdy \\
&= 0.
\end{aligned}$$

Therefore,

$$\begin{aligned}\oint_L \frac{-ydx+xdy}{x^2+y^2} &= \oint_{\partial B_\varepsilon(\mathbf{0})} \frac{-ydx+xdy}{x^2+y^2} \\ &= \int_0^{2\pi} \frac{-\varepsilon\sin\theta\cdot(-\varepsilon\sin\theta)+\varepsilon\sin\theta\cdot(\varepsilon\cos\theta)}{\varepsilon^2} d\theta \\ &= 2\pi.\end{aligned}$$

Example 3.5. Compute the line integral

$$\oint_L \sqrt{x^2+y^2}dx + y(xy+\log(x+\sqrt{x^2+y^2}))dy,$$

where L is the boundary of the region enclosed by $y^2 = x-1$ and $x = 2$ with anticlockwise orientation.

Example 3.6 (Green's formula)**.** Suppose D is enclosed by finitely many simple closed piecewisely C^1 curves. Let $\boldsymbol{n}$ be the unit normal vector field pointing outward of D. Suppose u, v are C^2, then

- [Green's first formula]

$$\oint_{\partial D} v\frac{\partial u}{\partial \boldsymbol{n}} ds = \iint_D (v\Delta u + \nabla u\cdot\nabla v)\, dxdy.$$

- [Green's second formula]

$$\oint_{\partial D} \left(v\frac{\partial u}{\partial \boldsymbol{n}} - u\frac{\partial v}{\partial \boldsymbol{n}}\right) ds = \iint_D (v\Delta u - u\Delta v)\, dxdy.$$

Proof. Let $\boldsymbol{n} = \cos\alpha\boldsymbol{i} + \sin\alpha\boldsymbol{j}$, then the unit tangent vector in the required direction is $\boldsymbol{\tau} = -\sin\alpha\boldsymbol{i} + \cos\alpha\boldsymbol{j}$, so we have

$$\frac{\partial u}{\partial \boldsymbol{n}} = \nabla u\cdot\boldsymbol{n} = \cos\alpha\frac{\partial u}{\partial x} + \sin\alpha\frac{\partial u}{\partial y} = \left(-\frac{\partial u}{\partial y}\boldsymbol{i} + \frac{\partial u}{\partial x}\boldsymbol{j}\right)\cdot\boldsymbol{\tau}.$$

As a consequence

$$\begin{aligned}\oint_{\partial D} v\frac{\partial u}{\partial \boldsymbol{n}} ds &= \oint_{\partial D} v\left(-\frac{\partial u}{\partial y}\boldsymbol{i} + \frac{\partial u}{\partial x}\boldsymbol{j}\right)\cdot\boldsymbol{\tau} ds \\ &= \oint_{\partial D} v\left(-\frac{\partial u}{\partial y}\boldsymbol{i} + \frac{\partial u}{\partial x}\boldsymbol{j}\right)\cdot d\boldsymbol{r} \\ &= \iint_D \left[\frac{\partial}{\partial x}\left(v\frac{\partial u}{\partial x}\right) + \frac{\partial}{\partial y}\left(v\frac{\partial u}{\partial y}\right)\right] dxdy \\ &= \iint_D (v\Delta u + \nabla u\cdot\nabla v)\, dxdy.\end{aligned}$$

□

4 Gauss divergence Theorem

Background: it originates from Gauss's study on the electric flux through a surface. Concretely speaking, if point charges q_i are placed at $P_i = (x_i, y_i, z_i)$ in $\mathbf{R}^3$ for $i = 1, 2, \cdots, n$, then what is the flux of electric field through a surface $S \subset \mathbf{R}^3$?

From empirical observation, $\boxed{\text{Coulumb's Law}}$[1] on the electric force between two statically placed point charges q_1 at $\boldsymbol{r}_1$ and q_2 at $\boldsymbol{r}_2$:

$$\boldsymbol{F} = k\frac{q_1 q_2}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|^3}(\boldsymbol{r}_1 - \boldsymbol{r}_2). \tag{4.1}$$

Together with *Superposition Principle* of electric field, Gauss derived the following $\boxed{\text{Gauss Law}}$:

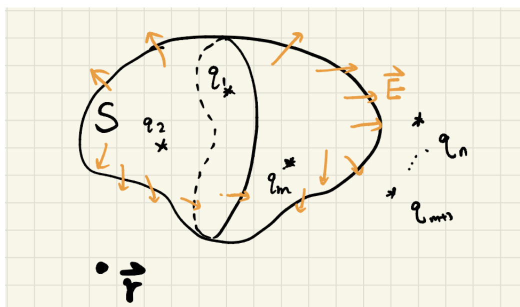


Figure 11: Gauss Law for electrostatics

"*The total electric flux through a closed surface S is proportional to the total charge inside the region enclosed by S. It neither depends on the charges outside the surface nor on the exact positions of the point charges.*"

The following mathematical theorem lies behind the derivative of Gauss's law:

Theorem 4.1 (Gauss divergence Theorem). *Suppose $\boldsymbol{v} = P(x,y,z)\boldsymbol{i} + Q(x,y,z)\boldsymbol{j} + R(x,y,z)\boldsymbol{k}$ is a C^1 vector field on $\overline{V}$ where V is enclosed by finitely many piecewisely C^1 surface. Then the total flux of $\boldsymbol{v}$ outside of ∂V is equal to the total divergence of $\boldsymbol{v}$ inside V. In terms of equation, it means*

$$\oint_{\partial V} P dy \wedge dz + Q dz \wedge dx + R dx \wedge dy = \iiint_V \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right) dxdydz$$

or

$$\oint_{\partial V} \boldsymbol{v} \cdot d\boldsymbol{S} = \iiint_V \nabla \cdot \boldsymbol{v} dV,$$

where ∂V is taken the outwards orientation.

[1] This law was actually discovered earlier and in a more accurate precision by Cavendish.

Notice that the "outwards orientation" on the boundary surface can be said as *induced orientation on the boundary* in the sense of the following:

Definition 4.2 (Induced orientation on the domain boundary)**.** Let V be a bounded domain in $\mathbf{R}^3$, suppose $\partial V = \sqcup_{i=1}^N S_i$, then the induced orientation on ∂V is the choice $\boldsymbol{n}_i$ on S_i such that $\boldsymbol{n}_i$ points outwards of V.

Proof. Similar to the treatment of Green's Theorem, let us first deal with region of simple shape.

Suppose $V = \{(x,y,z) \in \mathbf{R}^3 | (x,y) \in D \subset \mathbf{R}^2, z_1(x,y) \leq z \leq z_2(x,y)\}$ where z_1 and z_2 are two C^1 functions defined on D. Then

$$\begin{aligned}
\oint_{\partial V} R dx \wedge dy &= \iint_{S_0} + \iint_{S^+} + \iint_{S^-} R dx \wedge dy \\
&= \iint_{S_0} R\boldsymbol{k} \cdot \boldsymbol{n}_0 dS + \iint_D R(x,y,z_2(x,y)) dxdy - \iint_D R(x,y,z_1(x,y)) dxdy \\
&= \iint_D dxdy \int_{z_1(x,y)}^{z_2(x,y)} \frac{\partial R}{\partial z}(x,y,z) dz \\
&= \iiint_V \frac{\partial R}{\partial z} dxdydz.
\end{aligned}$$

Similarly, if V is of the shape $\{(x,y,z) \in \mathbf{R}^3 | (z,x) \in D, y_1(z,x) \leq y \leq y_2(z,x)\}$, then

$$\oint_{\partial V} Q dy \wedge dz = \iiint_V \frac{\partial Q}{\partial y} dxdydz$$

and if V is of the shape $\{(x,y,z) \in \mathbf{R}^3 | (y,z) \in D, x_1(y,z) \leq x \leq x_2(y,z)\}$ we can prove that

$$\oint_{\partial V} P dy \wedge dz = \iiint_V \frac{\partial P}{\partial x} dxdydz.$$

For the region in the statement of the theorem, we can use polygonal surface to approximate it. We can then divide the region into subregions of the types considered above. Actually each subregion can be treated in all of the three ways above and therefore

$$\begin{aligned}
&\oint_{\partial V_i} P dy \wedge dz + Q dz \wedge dx + R dx \wedge dy \\
&\qquad = \oint_{\partial V_i} P dy \wedge dz + \oint_{\partial V_i} Q dz \wedge dx + \oint_{\partial V_i} R dx \wedge dy \\
&\qquad = \iiint_{V_i} \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dxdydz.
\end{aligned}$$

Summing up all these and notice the cancellation on the intersecting surface in the interior of V, we conclude Gauss's Theorem.

□

4.1 Gauss's Law for electrostatics

We come back to the original motivation from electric field flux studied by Gauss. The electric field at $\boldsymbol{r}$ generated by the point charge q_i at $\boldsymbol{r}_i$ is

$$\boldsymbol{E}_i = k\frac{q_i}{|\boldsymbol{r}-\boldsymbol{r}_i|^3}(\boldsymbol{r}-\boldsymbol{r}_i),$$

and thus the overall field at $\boldsymbol{r}$ is

$$\boldsymbol{E} = \sum_{i=1}^{n}\boldsymbol{E}_i = k\sum_{i=1}^{n}\frac{q_i}{|\boldsymbol{r}-\boldsymbol{r}_i|^3}(\boldsymbol{r}-\boldsymbol{r}_i).$$

If $\boldsymbol{r}_i$ lies outside of S, then $\boldsymbol{E}_i$ is a smooth vector field in the interior region of S, and therefore

$$\begin{aligned}
\oint_S \boldsymbol{E}_i\cdot d\boldsymbol{S} &= kq_i\oint_S \frac{\boldsymbol{r}-\boldsymbol{r}_i}{|\boldsymbol{r}-\boldsymbol{r}_i|^3}\cdot d\boldsymbol{S}\\
&= kq_i\oint_S -\nabla\frac{1}{|\boldsymbol{r}-\boldsymbol{r}_i|}\cdot d\boldsymbol{S}\\
&= kq_i\iiint_V -\nabla\cdot\nabla\frac{1}{|\boldsymbol{r}-\boldsymbol{r}_i|}dV\\
&= -kq_i\iiint_V \Delta\frac{1}{|\boldsymbol{r}-\boldsymbol{r}_i|}dV.
\end{aligned}$$

Proposition 4.3. *The function* $\frac{1}{|\boldsymbol{r}-\boldsymbol{r}_i|}$ *is harmonic on* $\mathbf{R}^3\backslash\{\boldsymbol{r}_i\}$.

Proof. This is by straightforward computation. At $\boldsymbol{r}\neq\boldsymbol{r}_i$,

$$\begin{aligned}
\frac{\partial}{\partial x}\frac{1}{|\boldsymbol{r}-\boldsymbol{r}_i|} &= -\frac{1}{|\boldsymbol{r}-\boldsymbol{r}_i|^2}\frac{x-x_i}{|\boldsymbol{r}-\boldsymbol{r}_i|}\\
\frac{\partial^2}{\partial x^2}\frac{1}{|\boldsymbol{r}-\boldsymbol{r}_i|} &= -\frac{1}{|\boldsymbol{r}-\boldsymbol{r}_i|^3}+3\frac{(x-x_i)^2}{|\boldsymbol{r}-\boldsymbol{r}_i|^5}\\
\Delta\frac{1}{|\boldsymbol{r}-\boldsymbol{r}_i|} &= 0.
\end{aligned}$$

□

This implies that for $\boldsymbol{r}_i$ outside S, $\oint_S \boldsymbol{E}_i\cdot d\boldsymbol{S}=0$.

For $\boldsymbol{r}_i$ lying inside S, we cannot apply Gauss's Theorem to $\boldsymbol{E}_i$ since it is not a C^1 vector field inside S now. For any $\varepsilon>0$ sufficiently small,

$$\begin{aligned}
\oint_{\partial(V\backslash B(\boldsymbol{r}_i,\varepsilon))}\boldsymbol{E}_i\cdot d\boldsymbol{S} &= \oint_S \boldsymbol{E}_i\cdot d\boldsymbol{S} - \oint_{\partial B(\boldsymbol{r}_i,\varepsilon)}\boldsymbol{E}_i\cdot d\boldsymbol{S}\\
&= \iiint_{V\backslash B(\boldsymbol{r}_i,\varepsilon)}\Delta\frac{1}{|\boldsymbol{r}-\boldsymbol{r}_i|}dV\\
&= 0,
\end{aligned}$$

and therefore

$$\begin{aligned}\oint_S \boldsymbol{E}_i \cdot d\boldsymbol{S} &= \oint_{\partial B(\boldsymbol{r}_i,\varepsilon)} k\frac{q_i}{|\boldsymbol{r}-\boldsymbol{r}_i|^3}(\boldsymbol{r}-\boldsymbol{r}_i)\cdot d\boldsymbol{S} \\ &= 4\pi k q_i.\end{aligned}$$

By the linearity of surface integral and triple integral, together with the linearity of *divergence*, we get the *Gauss's Law*,

$$\oint_S \boldsymbol{E}\cdot d\boldsymbol{S} = \sum_{\boldsymbol{r}_i\in V} 4\pi k q_i = \frac{1}{\varepsilon_0}\sum_{\boldsymbol{r}_i\in V} q_i \tag{4.2}$$

in the integral formulation.

Remark 4.4. If Coulumb's Law does not satisfy the square reciprocal rule, then does Gauss's Law still hold?

4.2 The differential formulation of Gauss's Law on electrostatics and magnetic field

Suppose the charge density in $\mathbf{R}^3$ is given by ρ. Given a charge element ρdV inside V, it gives the electric field flux $\frac{\rho dV}{\varepsilon_0}$ through the surface S, and therefore by the *Superposition Principle* the total electric flux out of ∂V generated by all charges inside V is

$$\iiint_V \frac{\rho}{\varepsilon_0} dV = \oint_S \boldsymbol{E}\cdot d\boldsymbol{S}.$$

On the other hand, by Gauss's Theorem we have

$$\oint_S \boldsymbol{E}\cdot d\boldsymbol{S} = \iiint_V \nabla\cdot\boldsymbol{E}\, dV.$$

Since the region V is arbitrarily taken in $\mathbf{R}^3$, we get $\boxed{\text{Gauss's Law}}$ for electrostatics in the differential formulation:

$$\nabla\cdot\boldsymbol{E} = \frac{\rho}{\varepsilon_0}. \tag{4.3}$$

The point charge case is actually an ideal situation since there is no point charge in physic. The *electron cloud* means we cannot locate a charge at a particular position. If we introduce the notion *Dirac's delta function*, which is a *generalized function*:

$$\delta_{\boldsymbol{r}_i}(\boldsymbol{r}) = \begin{cases} \infty, & \boldsymbol{r}=\boldsymbol{r}_i \\ 0, & \boldsymbol{r}\neq\boldsymbol{r}_i \end{cases}$$

which satisfies the particular property:

$$\iiint_V \delta_{\boldsymbol{r}_i}(\boldsymbol{r})dV = \begin{cases} 0, & \boldsymbol{r}_i\notin V \\ 1, & \boldsymbol{r}_i\in V. \end{cases}$$

In terms of Dirac's delta function, for the situation we considered above, the Gauss's Law for point charge could be written as

$$\nabla \cdot \boldsymbol{E} = \frac{1}{\varepsilon_0} \sum_{i=1}^{n} q_i \delta_{\boldsymbol{r}_i}. \tag{4.4}$$

Gauss also discovered an law about (either natural magnet or electric current induced) magnetic field (this is observational fact, not first principle):

"*The magnetic flux of any magnetic field through a closed surface must be zero.*"

In term of integral, it means that

$$\oint_{\partial V} \boldsymbol{B} \cdot d\boldsymbol{S} = 0. \tag{4.5}$$

and in terms of differential equation, it means that

$$\nabla \cdot \boldsymbol{B} = 0 \tag{4.6}$$

which can be derived using Gauss's divergence Theorem. This law negates the existence of magnetic monopoles (磁单极子) , i.e. the magnetic analogy of point charge.

4.3 Examples

Example 4.5. Reconsider the example $\iint_S y(x-z)dy \wedge dz + x^2 dz \wedge dx + (y^2 + zx)dx \wedge dy$ where S is the boundary of the cuboid $0 \le x \le a, 0 \le y \le b, 0 \le z \le c$.

Solution. *The vector field* $\boldsymbol{v}$ *is* C^1 *on* $\mathbf{R}^3$*, and thus by Gauss's Theorem, it equal to*

$$\begin{aligned}
&\iiint_V \left(\frac{\partial}{\partial x} y(x-z) + \frac{\partial}{\partial y} x^2 + \frac{\partial}{\partial z}(y^2 + zx) \right) dxdydz \\
&= \iiint_V (x+y)\, dxdydz \\
&= \int_0^a xdx \int_0^b dy \int_0^c dz + \int_0^b ydy \int_0^a dx \int_0^c dz \\
&= \frac{1}{2} abc(a+b).
\end{aligned}$$

5 Stokes's Theorem

5.1 Physical laws about electric and magnetic fields

The various discoveries from the classical physics about electric and magnetic fields have far reached influence on our modern shape of the world. There are "deep" links between electric and magnetic field.

We've learned that electric current generates magnetic field by experiment.

$\boxed{\text{Oersted's Law}}$:

The magnetic field generated by a current I through an infinitely thin and long wire placed on z-axis is

$$\boldsymbol{B}(x, y, z) = \frac{\mu_0}{2\pi} \cdot \frac{-y\boldsymbol{i} + x\boldsymbol{j}}{x^2 + y^2}. \tag{5.1}$$

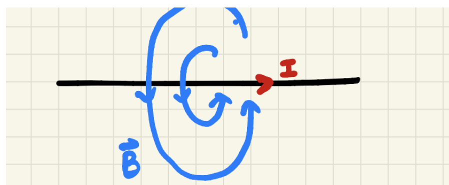


Figure 12: Oersted Law

The $\boxed{\text{Biot-Savart's Law}}$ we have encountered before:

$$d\boldsymbol{B} = \frac{\mu_0}{4\pi} \frac{I d\boldsymbol{l} \times \boldsymbol{r}}{r^3}. \tag{5.2}$$

There is also $\boxed{\text{Ampere's Right Hand Law}}$ for the magnetic field generated by electric current in the solenoid (螺线管)

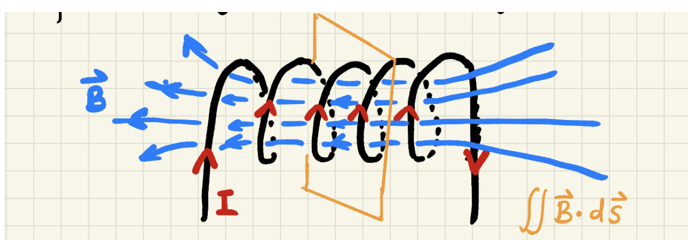


Figure 13: Ampere's Right Hand Rule

"*The circulation of the magnetic field along a closed loop is equal to the total current flux through a surface bounding this loop, for solenoid $B = \frac{\mu_0 N I}{l}$, moreover the direction of the current and the induced magnetism satisfies the right hand rule*".

Its integral form is

$$\oint_{\partial S} \boldsymbol{B} \cdot d\boldsymbol{r} = \mu_0 I \tag{5.3}$$

Another crucial development is $\boxed{\text{Faraday's law of electromagnetic induction}}$:

"The perpendicular movement of a magnet through the solenoid section generates circular electric field, which drives an electric current in the wire. The important *dynamo* is built upon this law. In terms of magnetic flux and induced electric field,

$$\frac{d}{dt}\iint_S \boldsymbol{B}\cdot d\boldsymbol{S} = -\oint_{\partial S}\boldsymbol{E}\cdot d\boldsymbol{r}, \tag{5.4}$$

where the "$-$" sign is determined by $\boxed{\text{Lenz's Rule}}$: the induced electric field in the solenoid generates magnetic field which is resistant to the input magnetic field，来拒去留。"

5.2 Stokes' Theorem

Theorem 5.1 (Stokes' Theorem)**.** *If $\boldsymbol{v} = P(x,y,z)\boldsymbol{i} + Q(x,y,z)\boldsymbol{j} + R(x,y,z)\boldsymbol{k}$ is a C^1 vector field in the domain V and $S \subset V$ is an oriented piecewise C^2 surface inside V, then*

$$\oint_{\partial S}\boldsymbol{v}\cdot d\boldsymbol{r} = \iint_S \nabla\times\boldsymbol{v}\cdot d\boldsymbol{S},$$

or

$$\begin{aligned}&\oint_{\partial S} Pdx + Qdy + Rdz\\ &\quad = \iint_S \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right) dy\wedge dz + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right) dz\wedge dx + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\wedge dy,\end{aligned}$$

where ∂S is equipped with the induced orientation on the surface boundary*.*

Definition 5.2 (Induced orientation on the surface boundary)**.** Let S be an oriented surface in $\mathbf{R}^3$ with the orientation given by the choice of unit normal vector field $\boldsymbol{n}$. Let $L = \partial S = \sqcup_{i=1}^N L_i$ be the boundary curve. The induced orientation on ∂L is the choice o_i on L_i such that $\boldsymbol{\tau}_i(p)\times\boldsymbol{n}(p)$ points outwards of S at $p \in L_i$.

This is equivalent to say, the outwards normal, the boundary tangential and the surface normal forms a right handed system.

Proof. The proof can be divided into several steps:

- Suppose S is parametrized by one coordinate patch $\boldsymbol{r} : (u,v) \in D(\subset \mathbf{R}^2) \to \mathbf{R}^3$ and $\boldsymbol{r}_u \times \boldsymbol{r}_v \cdot \boldsymbol{n} > 0$.

  Suppose the induced orientation on ∂S is represented by the unit tangential vector field $\boldsymbol{\tau}$, and let $\rho : [\alpha,\beta] \to \mathbf{R}^3$ given by $\rho(t) = \boldsymbol{r}(u(t), v(t))$ be

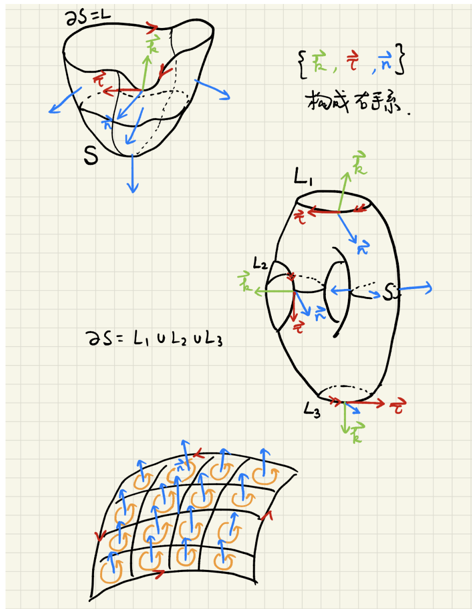


Figure 14: Boundary orientation and the decomposition of surface into small pieces

the positive arc length parameterization of ∂S, i.e. $\rho'(t) = \boldsymbol{\tau}(\rho(t))$. Here, $\eta : [\alpha, \beta] \to \mathbf{R}^2$ sending t to $\eta(t) = (u(t), v(t))$ is the parametric equation for ∂D.

- We claim that "the orientation of ∂D given by $\eta'(t)$ coincides with boundary orientation induced from the planar region D (the situation considered in Green's Theorem).

  Take $\eta'(t) = (u'(t), v'(t))$ and its clockwise $\frac{\pi}{2}$-rotated vector field $\boldsymbol{m} = (v'(t), -u'(t))$. By the above assumption,

$$
\begin{aligned}
d\boldsymbol{r}(\eta'(t)) &= \boldsymbol{r}_u u'(t) + \boldsymbol{r}_v v'(t) = \boldsymbol{\tau} \\
d\boldsymbol{r}(\boldsymbol{m}) &= \boldsymbol{r}_u v'(t) - \boldsymbol{r}_v u'(t),
\end{aligned}
$$

  and therefore

$$
\langle d\boldsymbol{r}(\boldsymbol{m}) \times d\boldsymbol{r}(\eta'), \boldsymbol{n} \rangle = \left[v'(t)^2 + u'(t)^2 \right] \langle \boldsymbol{r}_u \times \boldsymbol{r}_v, \boldsymbol{n} \rangle > 0.
$$

  This implies that $\{d\boldsymbol{r}(\boldsymbol{m}), \boldsymbol{\tau}, \boldsymbol{n}\}$ gives a right handed system of $\mathbf{R}^3$ and therefore $d\boldsymbol{r}(\boldsymbol{m})$ points outwards of S. This in turns tells that $\boldsymbol{m}$ points outwards of the planar region D, i.e. $\boldsymbol{m}$ is an out normal.

  As a conclusion, $\eta'(t)$ represents the boundary orientation of the planar region D.

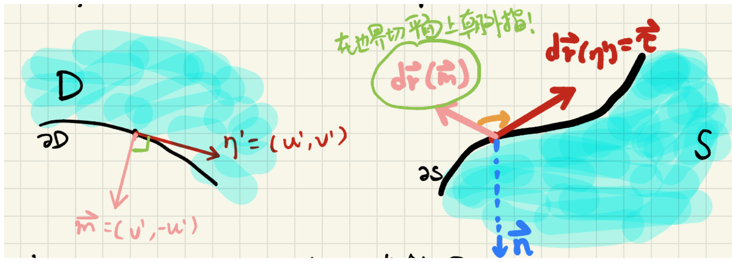


Figure 15: The induced boundary orientation from planar region and space domain are compatible

- We can now compute the line integral of $\boldsymbol{v}$ and surface integral of $\nabla \times \boldsymbol{v}$ in terms of the parametrization $\boldsymbol{r} \circ \eta(t)$ and $\boldsymbol{r}(u, v)$.

$$
\begin{aligned}
\oint_{\partial S} \boldsymbol{v} \cdot \boldsymbol{\tau} ds &= \int_{\alpha}^{\beta} \boldsymbol{v} \cdot \left(\boldsymbol{r}_u u'(t) + \boldsymbol{r}_v v'(t)\right) dt \\
&= \int_{\alpha}^{\beta} \Big[\left(\boldsymbol{v} \cdot \boldsymbol{r}_u\right) \boldsymbol{i} + \left(\boldsymbol{v} \cdot \boldsymbol{r}_v\right) \boldsymbol{j} \Big] \cdot \eta'(t) dt \\
&= \int_{\partial D} \Big[\left(\boldsymbol{v} \cdot \boldsymbol{r}_u\right) \boldsymbol{i} + \left(\boldsymbol{v} \cdot \boldsymbol{r}_v\right) \boldsymbol{j} \Big] \cdot d\boldsymbol{r} \\
&=_{\text{Green's Theorem}} \iint_D \left(\frac{\partial}{\partial u}(\boldsymbol{v} \cdot \boldsymbol{r}_v) - \frac{\partial}{\partial v}(\boldsymbol{v} \cdot \boldsymbol{r}_u) \right) du dv
\end{aligned}
$$

Notice that since $\boldsymbol{r}$ can be chosen to be C^2, $x_{uv} = x_{vu}$, etc, and thus

$$
\begin{aligned}
&\frac{\partial}{\partial u}(\boldsymbol{v} \cdot \boldsymbol{r}_v) - \frac{\partial}{\partial v}(\boldsymbol{v} \cdot \boldsymbol{r}_u) \\
&\qquad = \frac{\partial}{\partial u}\left(P x_v + Q y_v + R z_v\right) - \frac{\partial}{\partial v}\left(P x_u + Q y_u + R z_u\right) \\
&\qquad = \left(P_y y_u + P_z z_u\right) x_v + \left(Q_x x_u + Q_z z_u\right) y_v + \left(R_x x_u + R_y y_u\right) z_v \\
&\qquad\quad - \left(P_y y_v + P_z z_v\right) x_u - \left(Q_x x_v + Q_z z_v\right) y_u - \left(R_x x_v + R_y y_v\right) z_u \\
&\qquad = (R_y - Q_z)\frac{\partial(y, z)}{\partial(u, v)} + (P_z - R_x)\frac{\partial(z, x)}{\partial(u, v)} + (Q_x - P_y)\frac{\partial(x, y)}{\partial(u, v)} \\
&\qquad = \nabla \times \boldsymbol{v} \cdot (\boldsymbol{r}_u \times \boldsymbol{r}_v).
\end{aligned}
$$

Finally, we get that

$$
\oint_{\partial S} \boldsymbol{v} \cdot \boldsymbol{\tau} ds = \iint_S \nabla \times \boldsymbol{v} \cdot d\boldsymbol{S}.
$$

- For general surface, we can decompose $S = \cup_i S_i$ such that S_i and S_j has no overlapping interior and S_i can be parametrized by a single coordinate patch. Because of the cancellation of line integral in the interior of the surface S, we conclude Stokes's Theorem.

□

Green's Theorem can be treated as a special case of Stokes' Theorem, for which we view a planar region D as a surface in $\mathbf{R}^3$ with the "up" orientation. You may think *Gauss Theorem and Stokes' Theorem look very different*, however we will see later they can be unified formally.

5.3 Examples

Example 5.3. Let $S = \{(x, y, z)|x + y + z = 1, x, y, z \geq 0\}$ be the triangle region. Find the total work done by the force $\boldsymbol{F} = y^2\boldsymbol{i} + z^2\boldsymbol{j} + x^2\boldsymbol{k}$ along the circuit ∂S, with the clockwise orientation if we view it from the origin.

Solution. *If we equip S with the orientation $\boldsymbol{n} = \frac{1}{\sqrt{3}}(1, 1, 1)$, then the induced boundary orientation coincides with the given orientation of the boundary curve. Since the vector field and the surface are both smooth, we can apply Stokes' Theorem to compute the line integral:*

$$\begin{aligned}
\oint_{\partial S} \boldsymbol{F} \cdot d\boldsymbol{r} &= \iint_S \nabla \times \boldsymbol{F} \cdot d\boldsymbol{S} \\
&= \iint_S \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ \partial_x & \partial_y & \partial_z \\ y^2 & z^2 & x^2 \end{vmatrix} \cdot d\boldsymbol{S}. \\
&= \iint_S (-2z\boldsymbol{i} - 2x\boldsymbol{j} - 2y\boldsymbol{k}) \cdot \boldsymbol{n} dS \\
&= -\frac{2}{\sqrt{3}} \iint_S (z + x + y) dS \\
&= -\frac{2}{\sqrt{3}} Area(S) \\
&= -1.
\end{aligned}$$

For this problem, we can of course use the parametrization of ∂S to compute the integral directly.

$$\begin{aligned}
\oint_{\partial S} \boldsymbol{F} \cdot d\boldsymbol{r} &= \oint_{\partial S} y^2 dx + z^2 dy + x^2 dz \\
&= \int_{AB} y^2 dx + \int_{BC} z^2 dy + \int_{CA} x^2 dz \\
&= \int_1^0 (1 - x)^2 dx + \int_1^0 (1 - y)^2 dy + \int_1^0 (1 - z)^2 dz \\
&= -1.
\end{aligned}$$

Example 5.4. Compute $\oint_L (y^2 - z^2)dx + (2z^2 - x^2)dy + (3x^2 - y^2)dz$ where L is the intersecting curve of the plane $x + y + z = 2$ with the cylinder $|x| + |y| = 1$. Here L takes anticlockwise orientation if we view downwards this curve.

Solution.

$$
\begin{aligned}
&\oint_L (y^2-z^2)dx+(2z^2-x^2)dy+(3x^2-y^2)dz \\
&\qquad = \iint_S \left(\frac{\partial}{\partial y}(3x^2-y^2)-\frac{\partial}{\partial z}(2z^2-x^2)\right) dy\wedge dz \\
&\qquad\qquad + \left(\frac{\partial}{\partial z}(y^2-z^2)-\frac{\partial}{\partial x}(3x^2-y^2)\right) dz\wedge dx \\
&\qquad\qquad\qquad + \left(\frac{\partial}{\partial x}(2z^2-x^2)-\frac{\partial}{\partial y}(y^2-z^2)\right) dx\wedge dy \\
&\qquad = \iint_S (-2y-4z)dy\wedge dz+(-2z-6x)dz\wedge dx+(-2x-2y)dx\wedge dy \\
&\qquad = -2\iint_{|x|+|y|\le 1} (y+2(2-x-y))dy\wedge d(2-x-y) \\
&\qquad\qquad + (2-x-y+3x)d(2-x-y)\wedge dx+(x+y)dx\wedge dy \\
&\qquad = -2\iint_{|x|+|y|\le 1} (6+x-y)dxdy \\
&\qquad = -24.
\end{aligned}
$$

Example 5.5. Find the circulation of $\boldsymbol{v}=(y^2+z^2)\boldsymbol{i}+(z^2+x^2)\boldsymbol{j}+(x^2+y^2)\boldsymbol{k}$ along the curve L, which is the intersection of the sphere $x^2+y^2+z^2=R^2$ $(z\ge 0)$ and the cylinder $x^2+y^2=Rx$. Here L takes the anticlockwise orientation if we view the curve downwards.

Solution. *Take S to be the piece cut on the north hemisphere cut out by the cylinder, and the outward orientation. Then ∂S has the same orientation as L in the problem. We can then apply Stokes' Theorem*

$$
\begin{aligned}
\oint_L \boldsymbol{v}\cdot d\boldsymbol{r} &= \iint_S \nabla\times\boldsymbol{v}\cdot d\boldsymbol{S} \\
&= \iint_S \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ \partial_x & \partial_y & \partial_z \\ y^2+z^2 & z^2+x^2 & x^2+y^2 \end{vmatrix} \cdot d\boldsymbol{S} \\
&= 2\iint_S \Big((y-z)\boldsymbol{i}+(z-x)\boldsymbol{j}+(x-y)\boldsymbol{k}\Big)\cdot\frac{1}{R}\left(x\boldsymbol{i}+y\boldsymbol{j}+z\boldsymbol{k}\right)dS \\
&= 0.
\end{aligned}
$$

The last inequality is because $\nabla\times\boldsymbol{v}$ is perpendicular to the surface S everywhere. It is not very easy to compute directly the circulation of $\boldsymbol{v}$ via parametrization of the curve!

Example 5.6. Similarly, later we can prove that for a simple closed polytope $\partial\Omega=\cup_{i=1}^N\triangle A_iB_iC_i$ (the vertices are arranged such that $\overrightarrow{A_iB_i}\times\overrightarrow{A_iC_i}$ points outwards of Ω) with $A_i=(x_{i1},y_{i1},z_{i1}), B_i=(x_{i2},y_{i2},z_{i2}), C_i=(x_{i3},y_{i3},z_{i3})$, then

$$\mathrm{Vol}(\Omega) = \frac{1}{3}\sum_{i=1}^{N} \begin{vmatrix} x_{i1} & y_{i1} & z_{i1} \\ x_{i2} & y_{i2} & z_{i2} \\ x_{i3} & y_{i3} & z_{i3} \end{vmatrix}.$$

6 Conservative vector field and Divergence free vector field

6.1 Conservative fields/Curl free vector fields

Stokes' Theorem provides us a physical interpretation of "curl":

$$\nabla \times \boldsymbol{v}(\boldsymbol{x}) \cdot \boldsymbol{n} = \lim_{\varepsilon \to 0} \frac{1}{\pi \varepsilon^2} \oint_{\partial \mathbf{D}_{\boldsymbol{n}}(\boldsymbol{x},\varepsilon)} \boldsymbol{v} \cdot d\boldsymbol{r},$$

i.e. the projection of $\nabla \times \boldsymbol{v}$ in the direction $\boldsymbol{n}$ at $\boldsymbol{x}$ is given by the "(anti-clockwise) circulation density" of $\boldsymbol{v}$ along the infinitesimal circle normal to $\boldsymbol{n}$ and centered at $\boldsymbol{x}$, or in other words "the curl of a vector field measures the tendency for the vector field to swirl around", or "circulation per unit area".

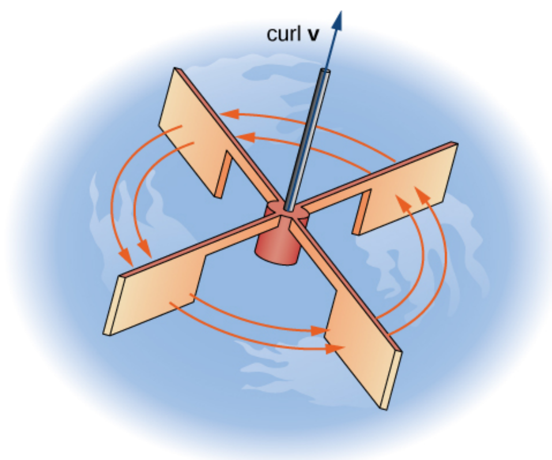


Figure 16: Physical interpretation of curl for $\boldsymbol{v}$: courtesy of cnx.org

The previous example about the work done by Newton's Gravity force shows that the work does not depend on the particular path connecting two points. Because of the similarity between gravity and electric force, the static electric field exhibits the same phenomenon:

- For any path L and L' connecting A, B,

$$\int_L \boldsymbol{v} \cdot d\boldsymbol{r} = \int_{L'} \boldsymbol{v} \cdot d\boldsymbol{r}.$$

- For any closed loop L,
$$\oint_L \boldsymbol{v} \cdot d\boldsymbol{r} = 0.$$

Definition 6.1 (Conservative vector field)**.** Let $\boldsymbol{v} = P\boldsymbol{i} + Q\boldsymbol{j} + Q\boldsymbol{k}$ be a continuous vector field on Ω, if for any $A, B \in \Omega$, the integral $\int_L \boldsymbol{v} \cdot d\boldsymbol{r} = \int_{L'} \boldsymbol{v} \cdot d\boldsymbol{r}$ for any piecewisely C^1 curves L, L' inside Ω connecting A, B (of course, the orientation of the curve is chosen such that it starts from A and ends at B), then we call $\boldsymbol{v}$ a *conservative vector field* in Ω.

What general properties does a conservative vector field posses? How to characterize a conservative vector field?

Theorem 6.2. *Suppose $\Omega \subset \mathbf{R}^3$ is a simply connected domain, and $\boldsymbol{v} = P\boldsymbol{i} + Q\boldsymbol{j} + R\boldsymbol{k}$ is a C^1 vector field in Ω, then the following are equivalent:*

1. *$\boldsymbol{v}$ is conservative;*
2. *There exists a C^2 function $\varphi : \Omega \to \mathbf{R}$ such that $\boldsymbol{v} = \nabla\varphi$, i.e. $Pdx + Qdy + Rdz = d\varphi$ (such function is called a potential function of $\boldsymbol{v}$);*
3. *The field is curve free, i.e. $\nabla \times \boldsymbol{v} = \mathbf{0}$.*

Remark 6.3.
1. If φ, ψ are both potential function for $\boldsymbol{v}$, then $\nabla(\varphi - \psi) = \mathbf{0}$. Since Ω is connected by assumption, we conclude that $\varphi \equiv \psi + C$ for a constant C; In practice, if we let φ take 0 at a particular point, for instance at ∞
2. If Ω is not simply connected, then potential function may not exist. For instance take $\Omega = \mathbf{R}^3 \backslash z-$ axis and $\boldsymbol{v} = \frac{-y\boldsymbol{i} + x\boldsymbol{j}}{x^2 + y^2}$ be the magnetic field generated by Oersted's line current: $\nabla \times \boldsymbol{v} = 0$ on Ω but
$$\oint_L \boldsymbol{v} \cdot d\boldsymbol{r} = 2\pi,$$
for L a horizontal circle centered on z-axis, therefore $\boldsymbol{v}$ has no (globally defined) potential function on Ω.

Proof. From 1 to 2: take $M_0 = (x_0, y_0, z_0) \in \Omega$, for any $M = (x, y, z) \in \Omega$ pick a C^1 curve L connecting M_0 and M, define

$$\varphi(M) := \int_L \boldsymbol{v} \cdot d\boldsymbol{r} = \int_L Pdx + Qdy + Rdz.$$

The condition in 1 tells us that φ is globally well-defined on Ω. Now, since

$$\begin{aligned}\frac{\partial\varphi}{\partial x}(x,y,z) &= \lim_{\Delta x\to 0}\frac{\varphi(x+\Delta x,y,z)-\varphi(x,y,z)}{\Delta x}\\ &= \lim_{\Delta x\to 0}\frac{1}{\Delta x}\int_{L_{[x,x+\Delta x]}} Pdx+Qdy+Rdz\\ &= \lim_{\Delta x\to 0}\frac{1}{\Delta x}\int_{x}^{x+\Delta x} P(t,y,z)dt\\ &= P(x,y,z).\end{aligned}$$

Similarly, $\frac{\partial\varphi}{\partial y}=Q$ and $\frac{\partial\varphi}{\partial z}=R$, i.e. $\boldsymbol{v}=\nabla\varphi$.

For item 2 to item 3: $\boldsymbol{v}=\nabla\varphi$, then by calculation $\nabla\times\boldsymbol{v}=\mathbf{0}$.

From item 3 to item 1: we use the unproven fact "In a simnply connected domain, any closed piecewise C^1 curve in Ω is the boundary of some orientable piecewise smooth C^1 surface (as the image of a C^1 maps from domain in $\mathbf{R}^2$ to $\mathbf{R}^3$)" [2]: then by Stokes Theorem,

$$\oint_L \boldsymbol{v}\cdot d\boldsymbol{r}=\iint_S \nabla\times\boldsymbol{v}\cdot d\boldsymbol{S}=0.$$

Theorem 6.4. *Let $\boldsymbol{v}$ be a C^1 vector field which is conservative and φ be one of its potential function, then for any $A,B\in\Omega$ and any piecewisely C^1 curve $L_{AB}\subset\Omega$ connecting A,B,*

$$\int_{L_{AB}}\boldsymbol{v}\cdot d\boldsymbol{r}=\varphi(B)-\varphi(A).$$

□

Example 6.5. Suppose the force field $\boldsymbol{F}$ is conservative, find the work done by $\boldsymbol{F}$.

Solution. *Let $-U$ be a potential function of $\boldsymbol{F}$, then we know that*

$$\int_{L_{AB}}\boldsymbol{F}\cdot d\boldsymbol{r}=U(A)-U(B).$$

Moreover, $\int_{L_{AB}}\boldsymbol{F}\cdot d\boldsymbol{r}=\int_{L_{AB}}m\ddot{\boldsymbol{r}}\cdot d\boldsymbol{r}=\int_\alpha^\beta m\ddot{\boldsymbol{r}}\cdot\dot{\boldsymbol{r}}dt=\int_\alpha^\beta\frac{d}{dt}\left(\frac{1}{2}m|\dot{\boldsymbol{r}}|^2\right)dt=\frac{1}{2}m|\boldsymbol{v}(B)|^2-\frac{1}{2}m|\boldsymbol{v}(A)|^2$, *i.e. the increased amount of kinetic energy is the decreased amount of potential energy.*

Example 6.6. Prove that $\boldsymbol{v}=(x^2-yz)\boldsymbol{i}+(y^2-zx)\boldsymbol{j}+(z^2-xy)\boldsymbol{k}$ has a potential function, and please find one of its potential function.

[2] This is probably nontrivial statement as can be seen for the case of the boundary curve of Mobius band and trefoil knot.

Solution. *First of all,* $\boldsymbol{v}$ *is* C^1*, we can compute*

$$\nabla \times \boldsymbol{v} = \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ \partial_x & \partial_y & \partial_z \\ x^2 - yz & y^2 - zx & z^2 - xy \end{vmatrix} = \mathbf{0}.$$

Since $\boldsymbol{v}$ *is defined on* $\mathbf{R}^3$ *which is obvious simply connected, it has a potential function. We can define*

$$\begin{aligned}\varphi(x, y, z) &= \int_{(0,0,0}^{(x,y,z)} (x^2 - yz)dx + (y^2 - zx)dy + (z^2 - xy)dz \\ &= \int_0^x x^2 dx + \int_0^y (y^2 - 0 \cdot x)dy + \int_0^z (z^2 - xy)dz \\ &= \frac{1}{3}x^3 + \frac{1}{3}y^3 + \frac{1}{3}z^3 - xyz.\end{aligned}$$

Notice that during the computation, we took the particular segments connecting $(0, 0, 0)$ *to* $(x, 0, 0)$, $(x, 0, 0)$ *to* $(x, y, 0)$ *and* $(x, y, 0)$ *to* (x, y, z).

Example 6.7. Calculate $\int_{(1,1,1)}^{(2,2,2)} \left(1 - \frac{1}{y} + \frac{y}{z}\right) dx + \left(\frac{x}{z} + \frac{x}{y^2}\right) dy - \frac{xy}{z^2} dz.$

Solution. *By computation, in the first octant,*

$$\begin{aligned}\nabla \times \boldsymbol{v} = &\left(\partial_y(-\frac{xy}{z^2}) - \partial_z(\frac{x}{z} + \frac{x}{y^2})\right) \boldsymbol{i} + \left(\partial_z(1 - \frac{1}{y} + \frac{y}{z}) - \partial_x(-\frac{xy}{z^2})\right) \boldsymbol{j} \\ &+ \left(\partial_x(\frac{x}{z} + \frac{x}{y^2}) - \partial_y(1 - \frac{1}{y} + \frac{y}{z}))\right) \boldsymbol{k} \\ = &\ \mathbf{0}.\end{aligned}$$

Since the first octant is simply connected, we can simply compute

$$\begin{aligned}\int_{(1,1,1)}^{(2,2,2)} \boldsymbol{v} \cdot d\boldsymbol{r} &= \int_1^2 \left(1 - \frac{1}{1} + \frac{1}{1}\right) dx + \int_1^2 \left(\frac{2}{1} + \frac{2}{y^2}\right) dy + \int_1^2 -\frac{4}{z^2} dz \\ &= 1 + 2 + \frac{2}{z}|_1^2 \\ &= 2.\end{aligned}$$

6.2 Divergence free vector field

Gauss's Theorem gives us a physical interpretation of "divergence":

$$\nabla \cdot \boldsymbol{v}(\boldsymbol{x}) = \frac{1}{\frac{4}{3}\pi\varepsilon^3} \oint_{\partial B(\boldsymbol{x},\varepsilon)} \boldsymbol{v} \cdot d\boldsymbol{S},$$

i.e. the divergence $\nabla \cdot \boldsymbol{v}$ at $\boldsymbol{x}$ is given by the "average (outwards) flux" through the infinitesimal sphere centered at $\boldsymbol{x}$, or in other words "flux per unit volume" in short.

In case $\boldsymbol{v}$ is the velocity field of a fluid, then $\nabla \cdot \boldsymbol{v} = \frac{1}{V}\frac{dV}{dt}$ where $V(t)$ measures the volume of fluid inside a fixed small closed surface at $\boldsymbol{x}$ at time t.

Definition 6.8. Suppose $\boldsymbol{v} = P\boldsymbol{i} + Q\boldsymbol{j} + R\boldsymbol{k}$ is a C^1 vector, if $\nabla \cdot \boldsymbol{v} = 0$, then we call $\boldsymbol{v}$ *divergence free*.

A typical example of divergence free vector field is magnetic field. If actually $\boldsymbol{v} = \nabla \times \boldsymbol{\alpha}$ for a C^2 vector field $\boldsymbol{\alpha}$, then $\nabla \cdot \boldsymbol{v} = 0$, i.e. a curl vector field is divergence free.

Definition 6.9. Let $\boldsymbol{v}$ be a C^1 vector field on a domain Ω, if there exists a vector field $\boldsymbol{v}$ on Ω such that $\boldsymbol{v} = \nabla \times \boldsymbol{\alpha}$, then we say $\boldsymbol{v}$ *admits vector potential*, and such $\boldsymbol{\alpha}$ is called *a vector potential* of $\boldsymbol{v}$.

Let $\boldsymbol{\alpha} = A\boldsymbol{i} + B\boldsymbol{j} + C\boldsymbol{k}$ and $\boldsymbol{v} = P\boldsymbol{i} + Q\boldsymbol{j} + R\boldsymbol{k}$, then $\nabla \times \boldsymbol{\alpha} = \boldsymbol{v}$ is equivalent to the following *system of first order PDE (partial differential equations)* :

$$\begin{cases} \frac{\partial C}{\partial y} - \frac{\partial B}{\partial z} & = P, \\ \frac{\partial A}{\partial z} - \frac{\partial C}{\partial x} & = Q, \\ \frac{\partial B}{\partial x} - \frac{\partial A}{\partial y} & = R. \end{cases} \tag{6.1}$$

The following questions need to be addressed:

1. Does the solution (A, B, C) exists?
2. Is solution unique?
3. How to find a (good) vector potential for a divergence free vector field?

6.3 Existence

About the existence: let us first try to find "solution of special form" (even though we do not know the existence yet). Let $C = 0$, then the system is reduced to

$$\begin{cases} \partial_z B = -P \\ \partial_z A = Q \\ \partial_x B - \partial_y A = R \end{cases}$$

By the first two equations, we can assume

$$A(x, y, z) = \int_{z_0}^{z} Q(x, y, t)dt$$

and

$$B(x, y, z) = -\int_{z_0}^{z} P(x, y, t)dt + f(x, y).$$

We plug these two terms into the third equation and obtain that

$$\frac{\partial f}{\partial x}(x, y) - \frac{\partial}{\partial x}\int_{z_0}^{z} P(x, y, t)dt - \frac{\partial}{\partial y}\int_{z_0}^{z} Q(x, y, t)dt = R(x, y, z),$$

i.e.

$$\frac{\partial f}{\partial x}(x,y) = R(x,y,z) + \int_{z_0}^{z} \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right)(x,y,t)dt.$$

The "divergence free" condition implies further that

$$\frac{\partial f}{\partial x}(x,y) = -\int_{z_0}^{z} \frac{\partial R}{\partial z}(x,y,t)dt + R(x,y,z) = R(x,y,z_0).$$

From this formula, we can obtain that

$$f(x,y) = \int_{x_0}^{x} R(s,y,z_0)ds.$$

Concluding the above argument, we can verify that

$$\begin{aligned} A &= \int_{z_0}^{z} Q(x,y,t)dt \\ B &= \int_{x_0}^{x} R(s,y,z_0)ds - \int_{z_0}^{z} P(x,y,t)dt \\ C &= 0 \end{aligned} \tag{6.2}$$

are all C^1 functions and $\boldsymbol{\alpha} = A\boldsymbol{i} + B\boldsymbol{k} + C\boldsymbol{k}$ satisfies $\nabla \times \boldsymbol{\alpha} = \boldsymbol{v}$.

Notice that in the above construction of A, B, C we've used that there exists $M_0 \in V$ such that the cuboid with diagonal M_0M lies in V for any $M \in V$. It the domain V satisfies this condition, then divergence free vector field in V admits a vector potential.

Theorem 6.10. *Let $\boldsymbol{v}$ be a C^1 vector field in the domain $V \subset \mathbf{R}^3$, then $\boldsymbol{v}$ is divergence free iff for any point $M \in V$ there exists a neighborhood of M on which $\boldsymbol{v}$ admits a vector potential.*

This is simply because any point M has a ball neighborhood whose shape satisfies the requirement in our above construction.

How about general domain in $\mathbf{R}^3$. Let us take the field $\boldsymbol{v} = \frac{\boldsymbol{r}}{r^3}$ defined on $\mathbf{R}^3 \backslash \{\mathbf{0}\}$ for example. Since $\boldsymbol{v} = -\nabla \frac{1}{r}$ we have $\nabla \cdot \boldsymbol{v} = \Delta \frac{1}{r} = 0$ on $\mathbf{R}^3 \backslash \{\mathbf{0}\}$. However,

$$\oint_{\partial B(\mathbf{0},1)} \boldsymbol{v} \cdot d\boldsymbol{S} = 4\pi.$$

If $\boldsymbol{v}$ admits a vector potential $\boldsymbol{\alpha}$, then $\oint_{\partial B(\mathbf{0},1)} \boldsymbol{v} \cdot d\boldsymbol{S} = \oint_{\partial B(\mathbf{0},1)} \nabla \times \boldsymbol{\alpha} \cdot d\boldsymbol{S} = 0$ where the last equality uses the Stokes's Theorem and the fact that $\partial B(\mathbf{0}, 1)$ has no boundary. This contradiction means that the electric field generated by a point charge does not admit a vector potential.

7 Differential form and generalized Stokes' Theorem

Let us first do a summary of the main things encountered in this chapter:

1.
$$\int_\gamma f ds = \int_\gamma f d\sigma_1;$$

2.
$$\int_S f dS = \int_S f d\sigma_2;$$

3. For an oriented curve γ, denoted by (γ, o) where o denotes its orientation:

$$\int_{(\gamma,o)} \boldsymbol{v} \cdot d\boldsymbol{r} = \int_{(\gamma,o)} Pdx + Qdy + Rdz;$$

   In case $\gamma = \partial S$ is piecewisely C^1 and $\boldsymbol{v}$ is C^1 on S, then there holds the *Stokes' Theorem*:

$$\oint_{(\gamma, o_S|_\gamma)} \boldsymbol{v} \cdot d\boldsymbol{r} = \iint_{(S,o_S)} \nabla \times \boldsymbol{v} \cdot d\boldsymbol{S}.$$

4. For an oriented surface S, denoted by (S, o_S), where o_S denotes its orientation and $\boldsymbol{v}$ is C^1 on S:

$$\int_{(S,o_S)} \boldsymbol{v} \cdot d\boldsymbol{S} = \int_{(S,o_S)} Pdy \wedge dz + Qdz \wedge dx + Rdx \wedge dy.$$

   In case $S = \partial V$ for some bounded domain and S is piecewisely C^1 and moreover $\boldsymbol{v}$ is C^1 on V, then there holds the *Gauss Theorem*:

$$\oint_{(S, o_V|_S)} \boldsymbol{v} \cdot d\boldsymbol{S} = \iint_V \nabla \cdot \boldsymbol{v} dxdydz,$$

   where $o_V|_S$ is the orientation on S induced from o on V.

We will see that Gauss Theorem and Stokes' Theorem in $\mathbf{R}^3$ about vector integral can be written in a more neat and uniform way:

Theorem 7.1 (Generalized Stokes' Theorem)**.** *Let Ω be an oriented k-dimensional submanifold of $\mathbf{R}^n$ with the orientation o_Ω, the induced orientation $o_\Omega|_{\partial\Omega}$ on the boundary $\partial\Omega$ is the one such that $o_\Omega|_{\partial\Omega} \cup \boldsymbol{N} = o_\Omega$ where $\boldsymbol{N}$ is an interior normal vector. Then for any C^1 k-form ω (even can be assumed only defined on Ω),*

$$\int_{\partial\Omega} \omega = \int_\Omega d\omega.$$

This is a deep duality between differential forms and chains, between exterior differentiation operator d and the boundary operator ∂!

8 Fourier series

8.1 Heat conduction

Consider the distribution of heat/temperature $T(x,y,z,t)$ in a space region $\Omega \subset \mathbf{R}^3$. It is reasonable to believe "*the flow of heat is resulted from the gradient/spatial difference of T*", which implies that "*the rate of change of the total amount of heat contained inside any surface is proportional to the total flux of heat outside of the surface*". Mathematically, this means that

$$\frac{d}{dt}\iiint_V T(x,y,z,t)dV = \frac{1}{k^2}\iint_{\partial V} \nabla T \cdot d\boldsymbol{S}. \tag{8.1}$$

By Divergence Theorem and the arbitrary choice of V inside Ω, we should have the Heat Equation (equation for heat conduction):

$$\Delta T = k^2 \frac{\partial T}{\partial t}. \tag{8.2}$$

When Fourier studied the *conduction of heat*, he considered the following model: let X be a straight stick placed on $[0,l]$ (very thin), if at time $t=0$ the temperature distribution is given by $f(x)$ and the temperature at the two endpoints are kept to be 0, then the temperature distribution $T(x,t)$ should satisfy

$$\begin{cases} \frac{\partial^2 T}{\partial x^2} = k^2 \frac{\partial T}{\partial t}, \\ T(0,t) = T(l,t) = 0, & t>0 \text{ (boundy condition);} \\ T(x,0) = f(x), & 0<x<l \text{ (initial condition).} \end{cases} \tag{8.3}$$

in which k is a constant depending on the material. How to solve this PDE?

Fourier used the familiar method "separating variables": assume $T(x,t) = \varphi(x)\psi(t)$ satisfies the equation, then

$$\varphi''(x)\psi(t) = k^2\varphi(x)\psi'(t),$$

or

$$\frac{\varphi''(x)}{k^2\varphi(x)} = \frac{\psi'(t)}{\psi(t)}.$$

In order this to hold, both sides must be equal to a constant $-\lambda$ (which is to be determined). For simplicity, we assume $k=1$, then we get two linear ODE (ordinary differential equation) of constant coefficients:

$$\begin{aligned} \varphi''(x) + \lambda\varphi(x) &= 0 \\ \psi'(t) + \lambda\psi(t) &= 0. \end{aligned} \tag{8.4}$$

Moreover, the boundary condition in (8.3) reasonably implies that $\varphi(0) = \varphi(l) = 0$.

- Solving the second equation: the equation implies that $e^{\lambda t}\psi(t) \equiv \psi(0)$ and therefore

$$\psi(t) = \psi(0)e^{-\lambda t}.$$

- Solving the first equation: if $\lambda \leq 0$, then the equation $\varphi''(x) = -\lambda\varphi$ subject to the boundary condition $\varphi(0) = \varphi(l) = 0$ must be identically 0 (think about why?). Therefore, we assume $\lambda > 0$. Notice that

$$\varphi''(x) + \lambda\varphi(x) = \left(\frac{d}{dx} + i\sqrt{\lambda}\right)\left(\frac{d}{dx} - i\sqrt{\lambda}\right)\varphi(x) = 0$$

implies that

$$\left(\frac{d}{dx} - i\sqrt{\lambda}\right)\varphi(x) = C_1 e^{-i\sqrt{\lambda}x}$$

for some complex number C_1. This means that

$$e^{i\sqrt{\lambda}x}\frac{d}{dx}\left(e^{-i\sqrt{\lambda}x}\varphi(x)\right) = C_i e^{-i\sqrt{\lambda}x},$$

from which we can integrate out to find

$$\varphi(x) = \tilde{C}_1 e^{i\sqrt{\lambda}x} + \tilde{C}_2 e^{-i\sqrt{\lambda}x}$$

for $\tilde{C}_1, \tilde{C}_2$ some undetermined complex constant. Notice that in the current situation we are only interested in real valued function, so we get the general form of real solution to $\varphi''(x) + \lambda\varphi(x) = 0$ is

$$\varphi(x) = C_1 \cos\sqrt{\lambda}x + C_2 \sin\sqrt{\lambda}x. \tag{8.5}$$

Taking into consideration $\varphi(0) = \varphi(l) = 0$, λ must satisfy the "quantization" condition

$$\sqrt{\lambda}l = n\pi, \quad n = 1, 2, \cdots, \tag{8.6}$$

In conclusion, we've derived a sequence of *separating variables* solution $T_n(x,t)$ of the heat equation subjecting to the boundary condition:

$$T_n(x,t) = e^{-\left(\frac{n\pi}{l}\right)^2 t} \sin\frac{n\pi}{l}x. \tag{8.7}$$

Because the original PDE and its boundary condition are both homogeneous (meaning all 0 in the current problem), the following formal "superposition" of the special solutions above

$$T(x,t) = \sum_{n=1}^{\infty} b_n e^{-\left(\frac{n\pi}{l}\right)^2 t} \sin\frac{n\pi}{l}x \tag{8.8}$$

is also a "solution" to the heat equation with zero boundary condition for any sequence b_n such that the convergence of the function series is "good".

The shape of the series do indicate that we must study the convergence of this function series....However, now we suppose everything goes as our wish. The final step is to satisfy the initial condtion $T(x, 0) = f(x)$, this amounts to say

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi}{l} x. \tag{8.9}$$

Is that possible to "expand" f defined on $[0, l]$ into such form of infinite series of sine functions of appropriate discrete set of frequencies? More generally, is it possible to expand f on $[0, l]$ as a series

$$f(x) = \frac{a_0}{2} + \sum_{i=1}^{\infty} \left(a_n \cos \frac{n\pi}{l} x + b_n \sin \frac{n\pi}{l} x \right) \tag{8.10}$$

for suitable constant a_n, b_n? The affirmative answer, then we can use the formula (8.8) to obtain "formal" solution to the (8.3).

Fourier himself believe this is true, however at the time many famous mathematicians (including Lagrange, Laplace and Legrendre) do not quite believe in it. Honestly speaking, it is a bit hard to believe why triogonometric functions are just right and enough to solve heat equation and to expand any functions!

8.2 Periodic function and orthogonality

Since each term in the series has period $2l$, we will look at if it is possible to expand periodic function in terms of trigonometric functions.

Definition 8.1. Let f be defined on $\mathbf{R}$, if there exists $T > 0$ such that $f(x + T) = f(x)$ for any $x \in \mathbf{R}$, then f is said to be periodic, and such a T is called a period of f.

Notice that if T is a period of f then nT is also for any $n \in \mathbf{N}$. One particularly important property for periodic function is that

$$\int_a^{a+T} f(x)dx = \int_0^T f(x)dx.$$

If g is a function of period T, then $f(x) = g(\frac{T}{2\pi} t)$ is a periodic function of period 2π. Therefore there is no loss in study functions with period 2π.

For a function defined on a closed interval $[a, b]$, there are several common ways of extending it to be a periodic function on $\mathbf{R}$:

- Central extension: we can first do a shift of the function, i.e. $g(x) := f(x + \frac{a+b}{2})$ with $x \in [-l, l]$ ($l = \frac{b-a}{2}$). Then we define

$$F(x) = \begin{cases} f(x - 2nl), & x \in \big((2n-1)l, (2n+1)l\big) \\ \frac{f(l)+f(-l)}{2}, & x = (2n+1)l \end{cases} .$$

- Even extension: we can shift the domain of f to $[0,l]$, i.e. $g(x)=f(x+a)$ with $x\in[0,l]$ $(l=b-a)$. Then, we first do an even extension

$$g_e(x)=\begin{cases} f(x), & x\in[0,l], \\ f(-x), & x\in[-l,0) \end{cases},$$

and then do periodic extension to a function on $\mathbf{R}$ with period $2l$.

- Odd extension: g is defined as above. Instead of extending g to be an even function, we extends it to be an odd function

$$g_o(x)=\begin{cases} f(x), & x\in(0,l], \\ 0, & x=0, \\ -f(-x), & x\in[-l,0) \end{cases},$$

and then we do a periodic extension of it to a function on $\mathbf{R}$ with period $2l$.

Definition 8.2. The collection of trigonometric functions

$$1,\cos x,\sin x,\cos 2x,\sin 2x,\cdots,\cos nx,\sin nx,\cdots \tag{$\dagger$}$$

is called trigonometric function system. They all have period 2π, and moreover they satisfies

$$\frac{1}{\pi}\int_{-\pi}^{\pi}\cos mx\cos nx=(1+\delta_{m0})\delta_{mn},\quad m,n=0,1,2,\cdots,$$
$$\frac{1}{\pi}\int_{-\pi}^{\pi}\sin mx\sin nx=0,\quad m,n=1,2,\cdots,$$
$$\frac{1}{\pi}\int_{-\pi}^{\pi}\cos mx\sin nx=0,\quad m=0,1,2,\cdots,n=1,2,\cdots.$$

Let $A[-\pi,\pi]=\{f:[-\pi,\pi]\to\mathbf{R}|f$ and f^2 are Riemann integrable on $[-\pi,\pi]\}$, and $\langle\cdot,\cdot\rangle_{L^2}$ be the pairing:

$$\langle f,g\rangle_{L^2}:=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)g(x)dx. \tag{8.11}$$

This pairing is bilinear and symmetric. However, it is not positive definite since $\langle f,f\rangle_{L^2}=0$ only implies "f is almost everywhere 0". Because of this, we can actually define an "equivalence relation" $\sim$ on $A[-\pi,\pi]$: $f\sim g$ iff $f-g$ is almost everywhere 0. The pairing naturally descends to the quotient space $A[-\pi,\pi]/\sim$ and on this quotient space it is positive definite.

If you do not feel very comfortable about "equivalence relation" and "quotient space", you can just confine yourself to $C[-\pi,\pi]$ and the L^2-pairing above is perfectly an inner product. Under this inner product, the trigonometric function system ($\dagger$) above is an *orthonormal* subset of $C[-\pi,\pi]$. Even in the finite dimensional, there are infinitely many orthonormal basis, the system ($\dagger$) is in no sense the only meaningful orthonormal set, but we will unarguably see that this system is a very important system.

8.3 The Fourier series of a function

Let us first do some heuristic argument to derive the definition of Fourier series. Suppose f is a periodic function with period 2π, and it can be written as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \tag{8.12}$$

formally. Suppose $f \in R[-\pi, \pi]$, then we get

$$\begin{aligned} \langle f(x), \cos nx \rangle_{L^2} &= a_n, \ n = 0, 1, 2, \cdots \\ \langle f(x), \sin nx \rangle_{L^2} &= b_n, \ n = 1, 2, \cdots . \end{aligned} \tag{8.13}$$

Thus, if f can be expanded as the form (8.12), it is reasonable to believe that the coefficient a_n, b_n take the above shape. By this reason, (8.13) are called the *Fourier coefficients* of f.

If f has a a singularity, to make sense of the coefficients, we must use the notion of *improper integral.* Therefore, we assume f is integrable and absolutely integrable in the next.

Definition 8.3. If f is integrable and absolutely integrable, then the series of trigonometric functions

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

where a_n, b_n are its Fourier coefficients is called the *Fourier series* of f. We denote this by

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) .$$

Theorem 8.4 (Dirichlet Theorem)**.** *Let f be a periodic function with period 2π.*

- *If f is piecewisely differentiable on any interval $[a, b]$, then the Fourier series of f converges on $\mathbf{R}$, and*

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) = \frac{f(x+0) + f(x-0)}{2}.$$

- *Assume f is $C^0(\mathbf{R})$ and piecewisely C^1, then the Fourier series of f converges to f uniformly on $\mathbf{R}$.*

In the statement, f is called piecewisely differentiable if for any $[a, b] \subset \mathbf{R}$, there exist a partition $a := t_0 < t_1 < \cdots < t_k = b$ such that

- For each i, $f : (t_{i-1}, t_i) \to \mathbf{R}$ is continuous;

- $f(t_{i-1}+0) = \lim_{t\to t_{i-1}^+} f(t), f(t_i - 0) = \lim_{t\to t_i^-} f(t)$ both exists;
- f is right differentiable at t_{i-1} and left differentiable at t_i in the sense that

$$f'_+(t_{i-1}) = \lim_{t\to 0^+} \frac{f(t) - f(t_{i-1}+0)}{t}, \quad f'_-(t_i) = \lim_{t\to 0^-} \frac{f(t) - f(t_i - 0)}{t}$$

  both exist.

This Theorem already lays a firm foundation for Fourier's theory on heat conduction since it deals with a quite large class of functions.

Example 8.5. Let f be a periodic function with period 2π, and it has the form

$$f(x) = \begin{cases} x, & 0 \le x < \pi, \\ 0, & -\pi \le x < 0. \end{cases}$$

Find its Fourier series.

Solution. *Let us first compute the Fourier coefficients by definition:*

$$\begin{aligned} a_0 &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx = \frac{\pi}{2}, \\ a_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx dx = \frac{1}{\pi}\int_0^{\pi} x\cos nx dx = \frac{(-1)^n - 1}{n^2\pi}, \\ b_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx dx = \frac{1}{\pi}\int_0^{\pi} x\sin nx dx = \frac{(-1)^{n+1}}{n}. \end{aligned}$$

Applying the above Dirichlet Theorem, we get

$$\frac{\pi}{4} + \sum_{n=1}^{\infty}\left[\frac{(-1)^n - 1}{n^2\pi}\cos nx + \frac{(-1)^{n+1}}{n}\sin nx\right] = \begin{cases} f(x), & x \ne (2k-1)\pi, \\ \frac{\pi}{2}, & x = (2k-1)\pi. \end{cases}$$

Plug $x = 0$, we obtain that

$$\sum_{k=1}^{\infty}\frac{1}{(2k-1)^2} = \frac{\pi^2}{8}.$$

As nonnegative series,

$$\sum_{n=1}^{\infty}\frac{1}{n^2} = \sum_{k=1}^{\infty}\frac{1}{(2k-1)^2} + \sum_{k=1}^{\infty}\frac{1}{(2k)^2} = \frac{\pi^2}{8} + \frac{1}{4}\sum_{n=1}^{\infty}\frac{1}{n^2}.$$

This implies the famous formula

$$\sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{\pi^2}{6}.$$

By the absolute convergence of the following alternating series, we obtain that

$$1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots = \sum_{k=1}^{\infty}\frac{1}{(2k-1)^2} - \sum_{k=1}^{\infty}\frac{1}{(2k)^2} = \frac{\pi^2}{12}.$$

8.4 Fourier expansion for function with general period $2l$

Let f be a periodic function with period $2l$, then $g(t) = f(\frac{l}{\pi}t)$ is a periodic function with period 2π. Then,

$$g(t) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nt + b_n \sin nt),$$

with

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} g(t)\cos nt = \frac{1}{\pi}\int_{-\pi}^{\pi} f\left(\frac{l}{\pi}t\right)\cos ntdt = \frac{1}{l}\int_{-l}^{l} f(x)\cos\frac{n\pi}{l}xdx$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} g(t)\sin ntdt = \frac{1}{\pi}\int_{-\pi}^{\pi} f\left(\frac{l}{\pi}t\right)\sin ntdt = \frac{1}{l}\int_{-l}^{l} f(x)\sin\frac{n\pi}{l}xdx.$$

This lead to the Fourier expansion of f as

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi}{l}x + b_n \sin\frac{n\pi}{l}x\right). \tag{8.14}$$

The Dirichlet Theorem also works for this expansion.

If f is only defined in the interval $[-l, l]$, we can do periodic extension of f as F on $\mathbf{R}$. Notice that F might be different from f at the point $x = \pm l$. However, this two point difference does not affect the integration, thus the Fourier coefficients of F can be simply computed using f on $[-l, l]$.

Theorem 8.6. *Let f be a piecewisely C^1 function on $[-l, l]$, then the Fourier series of f converges and*

$$\frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi}{l}x + b_n \sin\frac{n\pi}{l}x\right) = \begin{cases} \frac{f(x+0)+f(x-0)}{2}, & x \in (-l, l), \\ \frac{f(-l+0)+f(l-0)}{2}, & x = \pm l. \end{cases}$$

In particular, if moreover f is continuous on $[-l, l]$ with $f(-l) = f(l)$, then the Fourier series of f converges uniformly to f on $[-l, l]$.

If f is only defined in the interval $[a, b]$, how to expand f as reasonably good trigonometric series as above?

Let $g(t) = f(\frac{b+a}{2} + \frac{b-a}{2\pi}t)$, then g is defined on $[-\pi, \pi]$. Suppose its Fourier coefficients are $\tilde{a}_n, \tilde{b}_n$, then by the previous discussion

$$\begin{aligned}\tilde{a}_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} g(t)\cos ntdt = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\frac{b+a}{2} + \frac{b-a}{2\pi}t)\cos ntdt \\ &= \frac{1}{\pi}\int_{a}^{b} f(x)\cos\frac{2n\pi}{b-a}(x - x_0)\frac{2\pi}{b-a}dx \\ &= \frac{2}{b-a}\int_{a}^{b} f(x)\left(\cos\frac{2n\pi x_0}{b-a}\cos\frac{2n\pi}{b-a}x + \sin\frac{2n\pi x_0}{b-a}\sin\frac{2n\pi}{b-a}x\right)dx \\ &= a_n \cos\frac{2n\pi x_0}{b-a} + b_n \sin\frac{2n\pi x_0}{b-a},\end{aligned}$$

and similarly

$$\tilde{b}_n = -a_n \sin \frac{2n\pi x_0}{b-a} + b_n \cos \frac{2n\pi x_0}{b-a},$$

where

$$\begin{aligned} a_n &= \frac{2}{b-a} \int_a^b f(x) \cos \frac{2n\pi}{b-a} x dx, \\ b_n &= \frac{2}{b-a} \int_a^b f(x) \sin \frac{2n\pi x}{b-a} dx. \end{aligned}$$

According to

$$g(t) \sim \frac{\tilde{a}_0}{2} + \sum_{n=1}^{\infty} \left(\tilde{a}_n \cos nt + \tilde{b}_n \sin nt \right)$$

for $t \in [-\pi, \pi]$ and $f(x) = g\left(\frac{2\pi}{b-a}(x - x_0)\right)$, we obtain that

$$\begin{aligned} f(x) \sim \frac{\tilde{a}_0}{2} + \sum_{n=1}^{\infty} \Big[& \left(a_n \cos \frac{2n\pi x_0}{b-a} + b_n \sin \frac{2n\pi x_0}{b-a} \right) \cos \frac{2n\pi(x - x_0)}{b-a} \\ & + \left(-a_n \sin \frac{2n\pi x_0}{b-a} + b_n \cos \frac{2n\pi x_0}{b-a} \right) \sin \frac{2n\pi(x - x_0)}{b-a} \Big] \\ & = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{2n\pi}{b-a} x + b_n \sin \frac{2n\pi}{b-a} x \right). \end{aligned}$$

For f defined on $[a, b]$ we can shift it to be defined on $[0, b-a]$. If we do an even extension, and do Fourier expansion to the extended even and periodic function, we arrive at expanding f in terms of cosine function of the types $\cos \frac{n\pi}{b-a} x$ $(n = 0, 1, 2, \cdots)$. If we do an odd extension, and do Fourier expansion to the extended odd and periodic function, we arrive at expanding f in terms of sine functions of the type $\sin \frac{n\pi}{b-a} x$ $(n = 1, 2, \cdots)$.

8.5 Concrete Examples from Signal Processing

Example 8.7. Let $a \notin \mathbf{Z}$. Find the Fourier series of $f(x) = \cos ax$ on $[-\pi, \pi]$.

Solution. *We can periodically extend f as a function on $\mathbf{R}$ with period 2π which is continuous and piecewisely C^1. Since the function is even, we have*

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} \cos ax \cos nx dx \\ &= \frac{1}{\pi} \int_0^{\pi} [\cos(n+a)x + \cos(n-a)x] \, dx \\ &= \frac{(-1)^n}{\pi} \frac{2a \sin a\pi}{a^2 - n^2}, \ n = 0, 1, 2, \cdots. \\ b_n &= 0, \ n = 1, 2, \cdots. \end{aligned}$$

By Dirichlet Theorem, we have

$$\cos ax = \frac{\sin a\pi}{\pi}\left[\frac{1}{a} + \sum_{n=1}^{\infty}(-1)^n \frac{2a}{a^2-n^2}\cos nx\right],\ x \in [-\pi, \pi]. \tag{8.15}$$

Example 8.8 (Half-wave rectifier: 半波整流函数)**.** Consider the signal modeled on the following function on the interval $\left[-\frac{\pi}{\omega}, \frac{\pi}{\omega}\right]$:

$$f(t) = \begin{cases} 0, & -\frac{\pi}{\omega} \le t < 0, \\ E\sin\omega t, & 0 \le t \le \frac{\pi}{\omega}. \end{cases}$$

Find its Fourier series.

Solution. *Recall the formula for Fourier coefficients of function with period* $2l$*,*

$$\begin{aligned} a_n &= \frac{\omega}{\pi}\int_0^{\frac{\pi}{\omega}} E\sin\omega t\cos n\omega t dt \\ &= \frac{E\omega}{2\pi}\int_0^{\frac{\pi}{\omega}} \left[\sin(n+1)\omega t + \sin(n-1)\omega t\right] dt \\ &= \begin{cases} \frac{\left[(-1)^{n-1}-1\right]E}{(n^2-1)\pi}, & n \ne 1, \\ 0, & n = 1 \end{cases} \\ b_n &= \frac{\omega}{\pi}\int_0^{\frac{\pi}{\omega}} E\sin\omega t\sin n\omega t dt \\ &= \frac{E\omega}{2\pi}\int_0^{\frac{\pi}{\omega}} \left[\cos(n-1)\omega t - \cos(n+1)\omega t\right] dt \\ &= \begin{cases} 0, & n \ne 1, \\ \frac{E}{2}, & n = 1 \end{cases} \end{aligned}$$

By Dirichlet Theorem, we have

$$f(t) = \frac{E}{\pi} + \frac{E}{2}\sin\omega t + \frac{2E}{\pi}\sum_{k=1}^{\infty}\frac{1}{1-4k^2}\cos 2k\omega t,\ |t| \le \frac{\pi}{\omega}.$$

You may wonder why do we bother ourselves expanding these “simple” functions in terms of more complicated infinite series. One reason in signal processing is probably due to the fact that the sine and cosine signals (they are called *harmonics* 简谐波) are much easier to be generated, thus they can be used as the building blocks of much general signals.

In this example, the constant function $\frac{E}{\pi}$ is called the *direct current* component, the function $\frac{E}{2}\sin\omega t$ is called the *fundamental waveform*(or first harmonic). The component $a_n\cos\frac{n\pi}{l}x + b_n\sin\frac{n\pi}{l}x$ is called the *n-th harmonic*, with frequency $\omega_n = \frac{n\pi}{l}$. In practice, the amplitude of the n-th harmonic of a signal decays as n grows bigger, for example as $\frac{2E}{\pi}\frac{1}{4k^2-1}$ in our current example, thus making the tail of the series less and less important in shaping the whole signal.

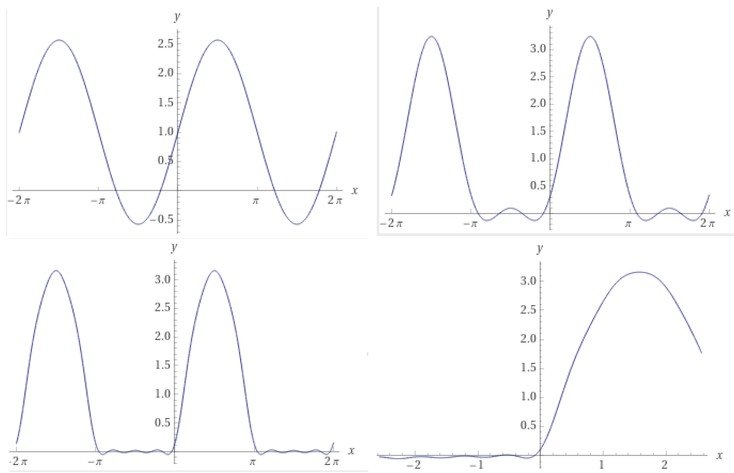


Figure 17: Partial sums (with $k = 0, 1, 3, 5$) of the Halfwave Fourier series

Example 8.9 (Rectangular Wave)**.** Find the Fourier series of the following function

$$f(x) = \begin{cases} \frac{1}{2h}, & |x| < h, \\ 0, & h \le |x| \le l. \end{cases}$$

Solution. *We periodic extends f to be defined on $\mathbf{R}$ and consider its Fourier series. Since f is even, $b_n \equiv 0$. And,*

$$a_n = \frac{1}{l}\int_{-h}^{h} \frac{1}{2h}\cos\frac{n\pi}{l}x dx = \frac{1}{n\pi h}\sin\frac{n\pi h}{l}, \ n = 1, 2, \cdots$$
$$a_0 = \frac{1}{l}.$$

By Dirichlet Theorem, we have

$$\frac{1}{2l} + \sum_{n=1}^{\infty}\frac{\sin\frac{n\pi h}{l}}{n\pi h}\cos\frac{n\pi}{l}x = \begin{cases} f(x), & |x| \le l, |x| \ne h, \\ \frac{1}{4h}, & |x| = h. \end{cases}$$

As a simple consequence, if we let $l = \pi, h = 1$, we conclude that

$$\sum_{n=1}^{\infty}\frac{\sin n}{n} = \frac{\pi - 1}{2} \tag{8.16}$$

Moreover, we can observe a pattern: let $l = 1$ and viewing h as a varying parameter, and look at the n-th Fourier coefficient

$$\frac{\sin n\pi h}{n\pi h},$$

which can be viewed as the frequency distribution of the original wave. It shows that "the smaller h is, the slower the amplitude decays" *or* "the more concentrated the signal is, the more sparsed the frequency distribution is".

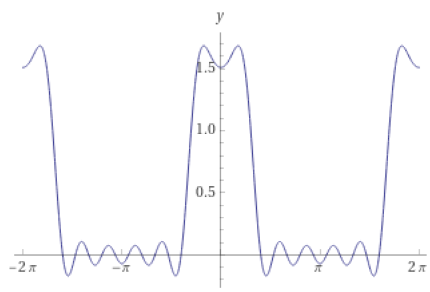


Figure 18: $S_7(x)$ (the first 7 frequencies of the Fourier series) for Squarewave with $h = 1$ and $l = \pi$

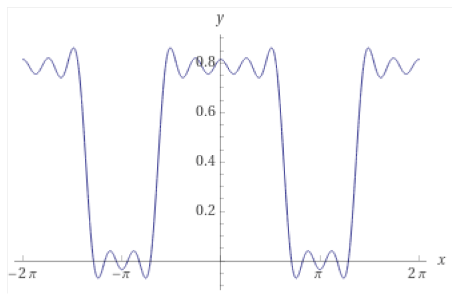


Figure 19: $S_7(x)$ (the first 7 frequencies of the Fourier series) for Squarewave with $h = 2$ and $l = \pi$

Example 8.10 (Triangluar wave)**.** Consider the function

$$f(x) = \begin{cases} x, & 0 \le x < \frac{l}{2}, \\ l - x, & \frac{l}{2} \le x \le l. \end{cases}$$

Expand it as a sine series and cosine series respectively.

Solution. • *To get a sine series, we can do odd extension. Then*

$$\begin{aligned} b_n &= \frac{2}{l}\int_0^l f(x) \sin\frac{n\pi}{l}x \\ &= \frac{2}{l}\int_0^{\frac{l}{2}} x \sin\frac{n\pi}{l}x dx + \frac{2}{l}\int_{\frac{l}{2}}^{l} (l-x)\sin\frac{n\pi}{l}x dx \\ &= \frac{2}{l}\int_0^{\frac{l}{2}} x\left[\sin\frac{n\pi x}{l} + \sin\frac{n\pi(l-x)}{l}\right] dx \\ &= \begin{cases} 0, & n = 2k, \\ \frac{(-1)^{k-1}4l}{(2k-1)^2\pi^2}, & n = 2k-1. \end{cases} \end{aligned}$$

By Dirichlet Theorem,

$$f(x) = \frac{4l}{\pi^2} \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{(2k-1)^2} \sin \frac{(2k-1)\pi x}{l}, \ 0 \le x \le l.$$

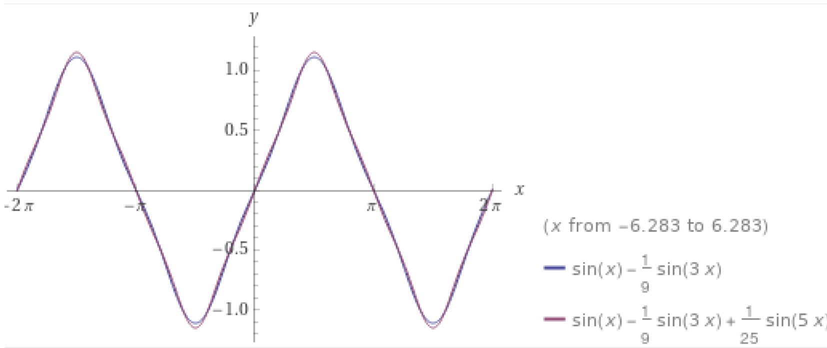


Figure 20: The partial sum of the sine Fourier series with $k = 2$ and $k = 3$

- *To get a cosine series for f, we do a even extension of f. Then $b_n \equiv 0$, and*

$$\begin{aligned}
a_n &= \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx \\
&= \frac{2}{l} \int_0^{\frac{l}{2}} x \cos \frac{n\pi x}{l} dx + \frac{2}{l} \int_{\frac{l}{2}}^{l} (l-x) \cos \frac{n\pi x}{l} d \\
&= \frac{2}{l} \int_0^{\frac{l}{2}} x \left[\cos \frac{n\pi x}{l} + \cos \frac{n\pi(l-x)}{l} \right] dx \\
&= \begin{cases} 0, & n = 2k-1, \\ \frac{4}{l} \int_0^{\frac{l}{2}} x \cos \frac{2k\pi x}{l} dx, & n = 2k. \end{cases}
\end{aligned}$$

And therefore all coefficient vanishes except $a_{4k+2} = -\frac{2l}{(2k+1)^2\pi^2}$ (with $k = 0, 1, 2, \cdots$) and $a_0 = \frac{l}{2}$. Finally, we obtain that

$$f(x) = \frac{l}{4} - \frac{2l}{\pi^2} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} \cos \frac{(4k+2)\pi}{l} x.$$

Example 8.11. Expand $f(x) = x^2$ on $[0, 1]$ as Fourier series.

Solution. *We can do even extension to get a Fourier expansion of f in terms*

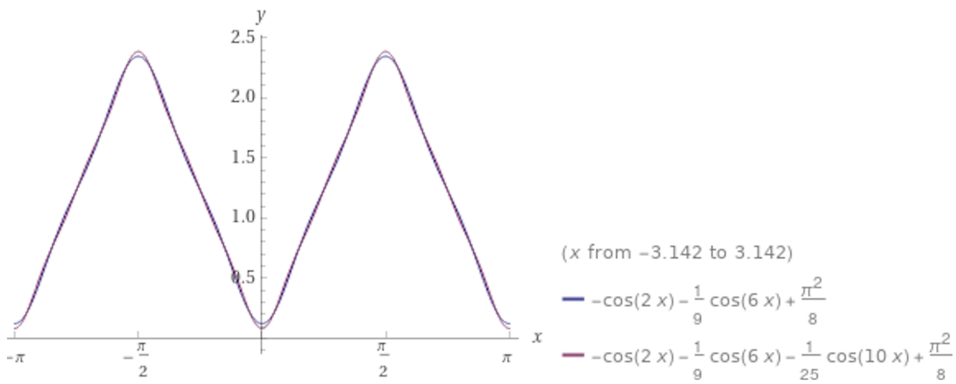


Figure 21: The partial sum of the cosine Fourier series with $k = 1$ and $k = 2$

of $\cos n\pi x$. *Concretely speaking,*

$$a_n = 2\int_0^1 x^2 \cos n\pi x dx = \frac{4(-1)^n}{n^2\pi^2},\ n = 1, 2, \cdots,$$
$$a_0 = \frac{2}{3}.$$

This leads to the formula

$$x^2 = \frac{1}{3} + \frac{4}{\pi^2}\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos n\pi x,\ 0 \le x \le 1.$$

We can also simply extend the function as a periodic function on $\mathbf{R}$ *with period* $2l = 1$, *then*

$$a_n = 2\int_0^1 f(x)\cos 2n\pi x dx = \frac{1}{n^2\pi^2},\ n = 1, 2, \cdots,$$
$$b_n = 2\int_0^1 f(x)\sin 2n\pi x dx = -\frac{1}{n\pi},\ n = 1, 2, \cdots,$$
$$a_0 = \frac{2}{3}.$$

This leads to that

$$x^2 = \frac{1}{3} + \sum_{n=1}^{\infty}\left(\frac{1}{n^2\pi^2}\cos 2n\pi x - \frac{1}{n\pi}\sin 2n\pi x\right),\ 0 < x < 1.$$

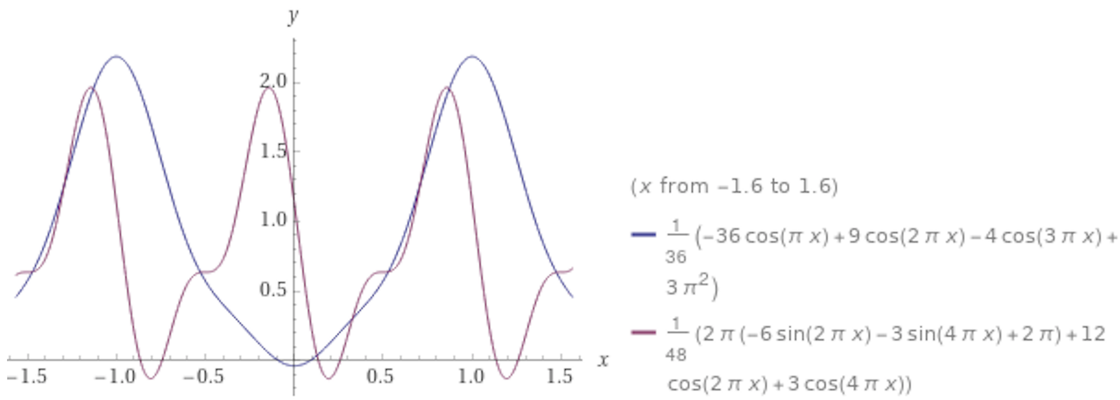


Figure 22: The partial sums for Cosine series (with $n = 3$ and Sine series (with $n = 2$) for $f(x) = \frac{\pi^2}{4}x^2$ on $[0, 1]$)

9 Convergence Theorems of Fourier series

9.1 Dirichlet Theorem: wonderful ideas of taking average

Let us look at the partial sums of Fourier series of a function f. Denote

$$S_N(f)(x) = \frac{a_0}{2} + \sum_{n=1}^{N} (a_n \cos nx + b_n \sin nx).$$

Let us rewrite this formula in a more transparent way:

$$\begin{aligned} S_N(f)(x) &= \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)dt + \sum_{n=1}^{N}\left(\frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos ntdt \cdot \cos nx + \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin ntdt \cdot \sin nx\right) \\ &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\left(\frac{1}{2} + \cos(x-t) + \cos 2(x-t) + \cdots + \cos N(x-t)\right) dt \\ &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x-t)\left(\frac{1}{2} + \cos t + \cos 2t + \cdots + \cos Nt\right) dt, \end{aligned}$$

where the last step is because the integrand is periodic function with period 2π. This formula for the partial sum of Fourier series inspires us the following definition:

Definition 9.1 (Dirichlet Kernal)**.** The sequence of functions $\{\mathcal{D}_N\}$, where

$$\mathcal{D}_N(x) = \frac{1}{2} + \cos x + \cdots + \cos Nx = \frac{\sin\left(N + \frac{1}{2}\right)x}{2\sin\frac{x}{2}},$$

is called the *Dirichlet Kernal*.

Under such terminology,

$$S_N(f)(x) = \int_{-\pi}^{\pi} f(x-t)\cdot \frac{1}{\pi}\mathcal{D}_N(t)dt. \tag{9.1}$$

Since

$$\int_{-\pi}^{\pi} \frac{1}{\pi}\mathcal{D}_N(t)dt = 1,\ N = 0, 1, 2, \cdots,$$

the function $\frac{1}{\pi}\mathcal{D}_N(t)$ can be viewed as a "weight function". Then, the value $S_N(f)(x)$ is the "weighted average" of f on the interval $[x-\pi, x+\pi]$, where each point $x-t$ takes the weight $\frac{1}{\pi}\mathcal{D}_N(t)$. Due to this explannation, if we can show that "the sequence of weight functions $\mathcal{D}_N$ concentrates more and more at $t=0$", then the contribution of the value $f(x-0)$ becomes more and more weighted among all possible values $f(x-t)$ as N grows bigger. This will lead to the desired consequence "$\lim_{N\to\infty} S_N(f)(x) = f(x)$"!

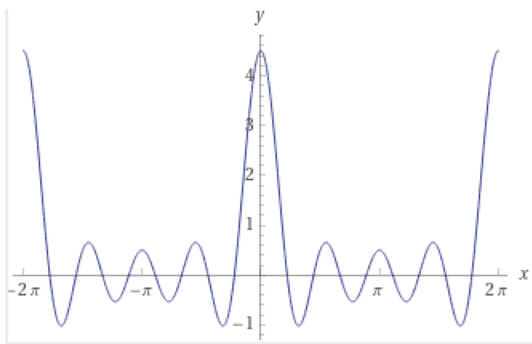


Figure 23: Dirichlet Kernal function $D_4(x)$ in two periods

However, the following peculiarity of Dirichlet Kernal/weight functions prevents life from becoming too easy. The weight at different points behaves remarkably differently:

- At $t=0$: the weight is $\frac{1}{\pi}\left(N+\frac{1}{2}\right)$;
- For nonzero rational multiple of π, i.e. $t = \frac{q}{p}\pi$, the "weight set" $\left\{\frac{1}{\pi}\frac{\sin\left(N+\frac{1}{2}\right)t}{2\sin\frac{t}{2}}\right\}_{N\in\mathbf{N}}$ is a finite set, in particular bounded but oscillates periodically with N;
- For general $t\in(0,\pi)$ such that $\frac{t}{\pi}$ is irrational, according to what we've shown in Mathematical Analysis I the "weight set" $\left\{\frac{1}{\pi}\frac{\sin\left(N+\frac{1}{2}\right)t}{2\sin\frac{t}{2}}\right\}_{N\in\mathbf{N}}$ is dense in $\left[-\frac{1}{\pi}\frac{1}{2\sin\frac{x}{2}}, \frac{1}{\pi}\frac{1}{2\sin\frac{x}{2}}\right]$.

This makes $\frac{1}{\pi}\mathcal{D}_N(x)$ diverges everywhere (as $N \to +\infty$)! Despite this peculiarity alluded above, we make the following observation: if f is "linear" near x, then it is still possible that the weighted average is very close to $f(x)$. This is the content of the deep theorem of Dirichlet mentioned previously, in which C^1 condition of f is assumed to amend the drawback of the kernal function.

Proof of Dirichlet Theorem. The difference of the partial sum of the Fourier series with the original function can be simplified as

$$\begin{aligned} S_N(f)(x) - f(x) &= \int_{-\pi}^{\pi} f(x-t) \cdot \frac{1}{\pi}\mathcal{D}_N(t)dt - 2\int_0^{\pi} f(x) \cdot \frac{1}{\pi}\mathcal{D}_N(t)dt \\ &= \int_0^{\pi} \Big(f(x+t) + f(x-t) - 2f(x)\Big) \cdot \frac{1}{\pi}\mathcal{D}_N(t)dt \\ &= \frac{1}{\pi}\int_0^{\pi} \frac{f(x+t)+f(x-t)-2f(x)}{2\sin\frac{t}{2}} \cdot \sin\left(N+\frac{1}{2}\right)t\ dt \\ &:= \frac{1}{\pi}\int_0^{\pi} \varphi(x,t)\sin\left(N+\frac{1}{2}\right)t\ dt \end{aligned}$$

For each fixed $x \in [-\pi, \pi]$, let us first investigate the property of the function $\varphi(x, \cdot)$ about $t \in (0, \pi]$.

- If f is assumed to be continuous on $\mathbf{R}$ and moreover f is left and right differentiable at x, then

  $$\lim_{t\to 0^+} \varphi(x,t) = \lim_{t\to 0} \frac{f(x+t) - f(x) - (f(x) - f(x-t))}{t} \cdot \frac{t}{2\sin\frac{t}{2}} = f'_+(x) - f'_-(x),$$

  i.e. $t = 0$ is a "removable singularity" of $\varphi(x,t)$. , then $\varphi(x,\cdot)$ is a continuous function of $t \in [0, \pi]$.

- If f is more generally only assumed to be piecewisely differentiable in the sense that for any $[a,b] \subset \mathbf{R}$, there exist a partition $a := t_0 < t_1 < \cdots < t_k = b$ such that

  - For each i, $f : (t_{i-1}, t_i) \to \mathbf{R}$ is continuous;
  - $f(t_{i-1}+0) = \lim_{t\to t_{i-1}^+} f(t), f(t_i - 0) = \lim_{t\to t_i^-} f(t)$ both exists;
  - f is right differentiable at t_{i-1} and left differentiable at t_i in the sense that

    $$f'_+(t_{i-1}) = \lim_{t\to 0^+} \frac{f(t) - f(t_{i-1}+0)}{t}, \quad f'_-(t_i) = \lim_{t\to 0^-} \frac{f(t) - f(t_i - 0)}{t}$$

    both exist,

  then instead of $S_N(f)(x) - f(x)$, we investigate

  $$S_N(f)(x) - \frac{f(x+0)+f(x-0)}{2} = \frac{1}{\pi}\int_0^{\pi} \varphi(x,t) \cdot \sin\left(N+\frac{1}{2}\right)t\ dt,$$

where

$$\varphi(x,t) = \frac{f(x+t) + f(x-t) - f(x+0) - f(x-0)}{2\sin\frac{t}{2}}.$$

Now, again $t = 0$ is a "removable singularity" of $\varphi(x,\cdot)$ and $\varphi(x,\cdot)$ is piecewisely continuous for $t \in [0,\pi]$.

In either of the above two cases, $\varphi(x,\cdot)$ is Riemann integrable about $t \in [0,\pi]$. By the following Riemann-Lebesgue Lemma, for all $t \in \mathbf{R}$,

$$\lim_{N\to\infty} S_N(f)(x) = \frac{f(x+0) + f(x-0)}{2}.$$

This finishes the proof of the first half of Dirichlet Theorem.

For item 2, the condition on f is: $f \in C^0(\mathbf{R})$, f is piecewisely C^1. Therefore, there exists a partition $T : -\pi = t_0 < t_1 < \cdots < t_k = \pi$ such that $f : [t_{i-1}, t_i] \to \mathbf{R}$ is C^1. Then, we have

$$\begin{aligned}
a_n &= \sum_{i=1}^{k} \frac{1}{\pi} \int_{t_{i-1}}^{t_i} f(x) \cos nx dx \\
&= \sum_{i=1}^{k} \left(\frac{f(x)\sin nx}{n\pi} |_{t_{i-1}}^{t_i} - \frac{1}{n\pi} \int_{t_{i-1}}^{t_i} f'(x) \sin nx dx \right) \\
&= -\frac{1}{n\pi} \int_{-\pi}^{\pi} f'(x) \sin nx dx \\
&= -\frac{b'_n}{n}. \\
b_n &= \sum_{i=1}^{k} \frac{1}{\pi} \int_{t_{i-1}}^{t_i} f(x) \sin nx dx \\
&= \sum_{i=1}^{k} \left(-\frac{f(x)\cos nx}{n\pi} |_{t_{i-1}}^{t_i} + \frac{1}{n\pi} \int_{t_{i-1}}^{t_i} f'(x) \cos nx dx \right) \\
&= \frac{1}{n\pi} \int_{-\pi}^{\pi} f'(x) \cos nx dx \\
&= \frac{a'_n}{n}.
\end{aligned}$$

It then follows that $2|a_n| \le \frac{1}{n^2} + b'^2_n$ and $2|b_n| \le \frac{1}{n^2} + a'^2_n$. According to Bessel's inequality (from next section)

$$\sum_{n=1}^{\infty} \left(a'^2_n + b'^2_n \right) \le \|f'\|^2_{L^2} < +\infty,$$

Weierstrass's M-test implies that

$$\frac{a_0}{2} \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

converges to f uniformly on $\mathbf{R}$. $\square$

Lemma 9.2 (Riemann-Lebesgue Lemma)**.** *Let $\varphi \in R[a,b]$, then*

$$\lim_{\lambda\to+\infty}\int_a^b \varphi(t)\cos\lambda t\ dt = \lim_{\lambda\to+\infty}\int_a^b \varphi(t)\sin\lambda t\ dt = 0.$$

Proof. For Darbourx Criteria for Riemann integrability, for any $\varepsilon > 0$, there exists a partition $T : a = t_0 < t_1 < \cdots < t_n = b$ such that

$$\sum_{i=1}^{n} \omega_i(\varphi)\Delta t_i < \varepsilon.$$

Define the following step function on $[a,b]$:

$$\psi(t) := \varphi(t_{i-1}),\ \ \forall t \in [t_{i-1}, t_i).$$

Then

$$\begin{aligned}
\left|\int_a^b \varphi(t)\cos\lambda t\ dt\right| &\le \left|\int_a^b \Big(\varphi(t)-\psi(t)\Big)\cos\lambda t\ dt\right| + \left|\int_a^b \psi(t)\cos\lambda t\ dt\right| \\
&\le \sum_{i=1}^{n}\int_{t_{i-1}}^{t_i}\sup_{[t_{i-1},t_i]}|\varphi(t)-\psi(t)|dt + \left|\sum_{i=1}^{n}\int_{t_{i-1}}^{t_i}\varphi(t_{i-1})\cos\lambda t\ dt\right| \\
&\le \sum_{i=1}^{n}\omega_i(\varphi)\Delta t_i + \sum_{i=1}^{n}\left|\frac{\varphi(t_{i-1})}{\lambda}\left(\sin\lambda t_i - \sin\lambda t_{i-1}\right)\right|.
\end{aligned}$$

Therefore, based on this partition T, there exists $M > 0$, such that for all $\lambda > M$,

$$\left|\int_a^b \varphi(t)\cos\lambda t\ dt\right| < 2\varepsilon.$$

$\square$

Remark 9.3. The Riemann-Lebesgue Lemma can be extended to the case where f has finitely many singularities such that the *improper integral* exists, for instance the function $\log\frac{1}{\sin x}$.

9.2 *Cesarò mean convergence*

The peculiar behavior of Dirichlet Kernal at $x \neq 0$ (i.e. the oscillating pattern for rational multiple of π, and the ergodic pattern for irrational multiple of π) indicates us if we arithmatically average the weights $\left\{\frac{\sin\left(N+\frac{1}{2}\right)x}{2\sin\frac{x}{2}}\right\}_{N=0,1,2,\cdots}$, then they will become smaller and smaller as $N \to +\infty$.

Definition 9.4 (Fejér Kernal)**.** The sequence of arithmetic average of Dirichlet Kernal, i.e. the sequence of functions $\{\mathcal{F}_N\}_{N=0,1,2,\cdots}$, where

$$\mathcal{F}_N(x) = \frac{\mathcal{D}_0(x) + \mathcal{D}_1(x) + \cdots + \mathcal{D}_N(x)}{N+1} = \frac{1}{2(N+1)} \frac{\sin^2\left(\frac{N+1}{2}x\right)}{\sin^2 \frac{x}{2}},$$

is called the *Fejér Kernal*.

In contrast to $\mathcal{D}_N$, the sequence $\mathcal{F}_N$ really concentrate more and more at $x = 0$, making the whole sequence behave more and more like a Dirac delta function at $x = 0$ as $N \to +\infty$.

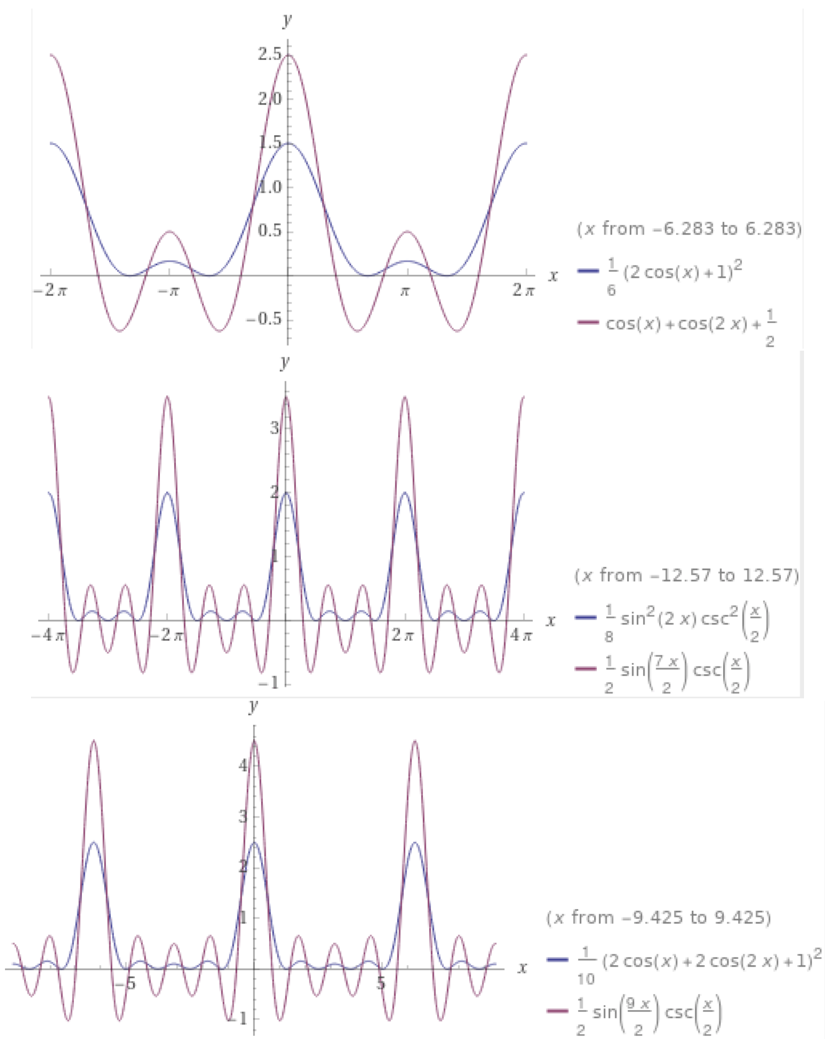


Figure 24: The graph of $\mathcal{D}_N$ (in purple) and $\mathcal{F}_N$ (in blue) with $N = 2, 3, 4$

Following this line, we define a function sequence by taking the arithmetic average of $\{S_N(f)\}_{N=0,1,2,\cdots}$, i.e. we form $\{\sigma_N(f)\}_{N=0,1,2,\cdots}$ where

$$\sigma_N(f)(x) = \frac{S_0(f)(x) + S_1(f)(x) + \cdots + S_N(f)(x)}{N+1}.$$

This is called the *Cesarò mean* of f. Notice that as $S_N(f)$, i.e. it is also a trigonometric series of degree no bigger than N.

Fix any point x, Stolz's Theorem tells that as long as $S_N(f)(x)$ converges to a, then $\sigma_N(f)(x)$ converges to a. The example $\{(-1)^n\}$ shows that in general $\sigma_N(f)(x)$ may converge at a point even though $\{S_N(f)(x)\}$ does not converge. According to the formula

$$S_N(f)(x) = \int_{-\pi}^{\pi} f(x-t) \cdot \frac{1}{\pi}\mathcal{D}_N(t)dt,$$

we have

$$\sigma_N(f)(x) = \int_{-\pi}^{\pi} f(x-t) \cdot \frac{1}{\pi}\mathcal{F}_N(t)dt. \tag{9.2}$$

Following the idea of the proof of Dirichlet Theorem viewing $S_N(f)$ as a "weighted average" of f using the Dirichlet Kernal/weight function $\frac{1}{\pi}\mathcal{D}_N$, in the current situation $\sigma_N(f)$ can be viewed as a "weighted average" of f using (better behaved) Fejér Kernal/weight function $\frac{1}{\pi}\mathcal{F}_N$. We state without proof the following:

Theorem 9.5 (Fejér Theorem)**.** *If $f \in C^0(\mathbf{R})$ has period 2π, then $\sigma_N(f)$ converges to f uniformly on $\mathbf{R}$ as $N \to +\infty$.*

Question 9.6. *Can you do average to Fejér kernal to obtain better behaved kernal functions?*

10 Squared mean convergence

In this section, we study the convergence of Fourier series in a *more abstract* way, i.e. from the viewpoint of function space (of infinite dimension, equipped with suitable inner product or norm).

We define an L^2 pairing for $f, g \in R[-\pi, \pi]$:

$$\langle f, g\rangle_{L^2} = \int_{-\pi}^{\pi} f(x)g(x)dx.$$

We've seen before that under this L^2 pairing, the following system

$$\left\{\frac{1}{\sqrt{2\pi}}, \frac{\cos x}{\sqrt{\pi}}, \frac{\sin x}{\sqrt{\pi}}, \frac{\cos 2x}{\sqrt{\pi}}, \frac{\sin 2x}{\sqrt{\pi}}, \cdots, \right\}$$

is an orthonormal subset of $R[-\pi, \pi]$.

10.1 Square mean convergence

Theorem 10.1. *Let $V_n = \mathrm{Span}\{1, \cos x, \sin x, \cdots, \cos nx, \sin nx\}$ be the finite dimensional subspace of $R[-\pi, \pi]$* [3]*. For any $f \in R[-\pi, \pi]$, denote $S_n(f)$ as the partial sum of its Fourier series above, then for any $\psi \in V_n$,*

$$\|f - \psi\|_{L^2}^2 = \|f - S_n(f)\|_{L^2}^2 + \|S_n(f) - \psi\|_{L^2}^2.$$

1. *The Fourier series $S_n(f)$ is the unique point in V_n which is closest to f under the L^2-distance, i.e. $S_n(f) = \mathrm{pr}_{V_n}(f)$;*
2. *[Bessel's inequality]*

$$\frac{a_0^2}{2} + \sum_{n=1}^{\infty} \left(a_n^2 + b_n^2\right) \leq \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)^2 dx.$$

Proof. Since $\psi \in V_n$, we can write is as

$$\psi(x) = \frac{\alpha_0}{2} + \sum_{k=1}^{n} \left(\alpha_k \cos kx + \beta_k \sin kx\right).$$

We have

$$\begin{aligned}
\|f - \psi\|_{L^2}^2 &= \|(f - S_n(f)) + (S_n(f) - \psi)\|_{L^2}^2 \\
&= \|f - S_n(f)\|_{L^2}^2 + \|S_n(f) - \psi\|_{L^2}^2 \\
&\qquad - 2\langle f - S_n(f), \frac{a_0 - \alpha_0}{2} + \sum_{k=1}^{n} \left((a_k - \alpha_k) \cos kx + (b_k - \beta_k) \sin kx\right)\rangle_{L^2} \\
&= \|f - S_n(f)\|_{L^2}^2 + \|S_n(f) - \psi\|_{L^2}^2,
\end{aligned}$$

where the last equality holds because

$$\begin{aligned}
\langle f - S_n(f), \cos kx\rangle_{L^2} &= \langle f, \cos kx\rangle_{L^2} - a_k = 0,\ k = 0, 1, \cdots, n; \\
\langle f - S_n(f), \sin kx\rangle_{L^2} &= \langle f, \sin kx\rangle_{L^2} - b_k = 0,\ k = 1, 2, \cdots, n.
\end{aligned}$$

The consequence is that

$$\|f - \psi\|_{L^2}^2 \geq \|f - S_n(f)\|_{L^2}^2,$$

and the equality is realized iff $\psi = S_n(f)$ since

$$\|S_n(f) - \psi\|_{L^2}^2 = \pi \left[\frac{(a_0 - \alpha_0)^2}{2} + \sum_{k=1}^{n} \Big((a_k - \alpha_k)^2 + (b_k - \beta_k)^2 \Big)\right].$$

If we take $\psi = 0$ above, then we obtain

$$\|f\|_{L^2}^2 \geq \|S_n(f)\|_{L^2}^2$$

[3] We restrict ourselves to Riemann integrable functions at this moment.

for any $n \in \mathbf{N}$. Since the left is a finite number, the series $\pi \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} \left(a_n^2 + b_n^2\right)\right]$ is bounded from above and thus converges. Moreover, its sum is not bigger than $\|f\|_{L^2}^2$, which establishes the Bessel's inequality. □

A very natural question is whether Bessel's inequality is actually an equality?

Theorem 10.2 (Parseval equality (1806)/Square mean convergence (平方平均收敛)). *For any $f \in R[-\pi, \pi]$, it holds that*

$$\frac{a_0^2}{2} + \sum_{n=1}^{\infty} \left(a_n^2 + b_n^2\right) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)^2 dx.$$

This is equivalent to say that the function sequence $S_n(f)$ converges to f in square mean sense, i.e. $\lim_{n\to\infty} \|S_n(f) - f\|_{L^2} = 0$.

Proof. Since $f \in R[-\pi, \pi]$, we have $M > 0$ such that $|f| \leq M$ on $[-\pi, \pi]$. Moreover, for any $\varepsilon > 0$, there exists a partition $T : -\pi = t_0 < t_1 < \cdots < t_n = \pi$ such that $\|T\| < \varepsilon$ and

$$\sum_{i=1}^{n} \omega_i(f) \Delta t_i < \varepsilon.$$

Let φ be a piecewisely linear function, which connecting $(t_{i-1}, f(t_{i-1}))$ with $(t_i, f(t_i))$ for $i = 1, 2, \cdots, n-1$, and connecting $(t_{n-1}, f(t_{n-1}))$ with $(t_n, f(t_0)) = (b, f(a))$. This function is an "approximating" function of f in the sense that

$$\begin{aligned}
\|f - \varphi\|_{L^2}^2 &= \sum_{i=1}^{n} \frac{1}{\pi} \int_{t_{i-1}}^{t_i} (f(t) - \varphi(t))^2 dt \\
&\leq \sum_{i=1}^{n-1} \omega_i(f)^2 \Delta t_i + 4M^2 \Delta t_n \\
&\leq (2M + 4M^2)\varepsilon.
\end{aligned}$$

Since $\varphi(-\pi) = \varphi(\pi)$, it can be periodically extended as a continuous function on $\mathbf{R}$ which has period 2π and is piecewisely C^1. We can apply item 2 of Dirichlet's Theorem, there exists $N \in \mathbf{N}_+$ such that for any $n > N$,

$$\|\varphi - S_n(\varphi)\|_{L^2}^2 < \varepsilon$$

by the uniform convergence. By the triangle inequality,

$$\begin{aligned}
\|f - S_n(f)\|_{L^2}^2 &\leq \|f - S_n(\varphi)\|_{L^2}^2 \leq 2\|f - \varphi\|_{L^2}^2 + 2\|\varphi - S_n(\varphi)\|_{L^2}^2 \\
&\leq 2(2M + 4M^2)\varepsilon + 2\varepsilon.
\end{aligned}$$

By previous theorem, $\lim_{n\to+\infty} \left(\|f\|_{L^2}^2 - \|S_n(f)\|_{L^2}^2\right) = 0$. This is precisely the content of Parseval equality. □

Corollary 10.3 (Denseness of the trigonometric system)**.** *Let $f \in C[-\pi, \pi]$ be such that f is orthogonal to $\{1, \cos x, \sin x, \cos 2x, \sin 2x, \cdots\}$, then $f(x) \equiv 0$. In particular, two continuous functions with the same set of Fourier coefficients must be equal.*

Corollary 10.4 (L^2-inner product preserving)**.** *Let $f, g \in R[-\pi, \pi]$ be two real-valued functions, and let $\{a_n, b_n\}, \{\tilde{a}_n, \tilde{b}_n\}$ be the Fourier coefficients of f, g respectively, then*

$$\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)g(x)dx = \frac{a_0\tilde{a}_0}{2} + \sum_{n=1}^{\infty}\left(a_n\tilde{a}_n + b_n\tilde{b}_n\right)$$

We can compare the condition of this statement with the stringent condition on integration term by term in power series.

Proof. Notice that $\{a_n \pm \tilde{a}_n\}, \{b_n \pm \tilde{b}_n\}$ are the Fourier coefficients of $f \pm g \in R[-\pi, \pi]$. Apply Parseval equality to $f + g$ and $f - g$ respectively. The claimed equality holds since we can do subtraction term by term for two absolutely converging sequences. □

Corollary 10.5 (Integration by terms)**.** *Let $f \in R[-\pi, \pi]$ such that its Fourier series is*

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos nx + b_n \sin nx\right),$$

then for any $[a, b] \subset [-\pi, \pi]$, there exists

$$\int_a^b f(x)dx = \int_a^b \frac{a_0}{2}dx + \sum_{n=1}^{\infty}\int_a^b \left(a_n \cos nx + b_n \sin nx\right)dx.$$

Proof. Take $g(x) = \chi_{[a,b]}(x)$, i.e. the characteristic function of the interval $[a, b]$. The Fourier coefficients of g is

$$\begin{aligned}
\tilde{a}_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} g(x)\cos nx dx \\
&= \frac{1}{\pi}\int_a^b \cos nx dx \\
\tilde{b}_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} g(x)\sin nx dx \\
&= \frac{1}{\pi}\int_a^b \sin nx dx.
\end{aligned}$$

Therefore, according to last corollary of Parseval equality:

$$
\begin{aligned}
\int_a^b f(x)dx &= \int_{-\pi}^{\pi} f(x)g(x)dx \\
&= \pi\Big\{\frac{a_0\tilde{a}_0}{2} + \sum_{n=1}^{\infty}\Big(a_n\tilde{a}_n + b_n\tilde{b}_n\Big)\Big\} \\
&= \int_a^b \frac{a_0}{2}dx + \sum_{n=1}^{\infty}\left(\int_a^b a_n \cos nx dx + \int_a^b b_n \sin nx dx\right)
\end{aligned}
$$

which is the *integration by terms* formula. □

10.2 From $R[-\pi,\pi]$ to $L^2[-\pi,\pi]$

Denote ℓ^2 the space of square summable sequence $\{a_0, a_1, b_1, a_2, b_2, \cdots\}$, i.e.

$$
\frac{a_0^2}{2} + \sum_{n=1}^{\infty}\left(a_n^2 + b_n^2\right) < +\infty.
$$

And we endow ℓ^2 with the following inner product: for $\alpha = \{a_0, a_1, b_1, a_2, b_2, \cdots\}$, $\widetilde{\alpha} = \{\widetilde{a}_0, \widetilde{a}_1, \widetilde{b}_1, \widetilde{a}_2, \widetilde{b}_2, \cdots,\}$ we define

$$
\langle \alpha, \widetilde{\alpha}\rangle_{\ell^2} = \pi\left(\frac{a_0\widetilde{a}_0}{2} + \sum_{n=1}^{\infty}\left(a_n\widetilde{a}_n + b_n\widetilde{b}_n\right)\right).
$$

Parseval equality says that the following map:

$$
\begin{aligned}
\mathcal{F} : C[-\pi,\pi] &\longrightarrow \ell^2 \\
f &\mapsto \{a_0, a_1, b_1, a_2, b_2, \cdots\}
\end{aligned}
\tag{10.1}
$$

sending a continuous function to its Fourier coefficient sequence is an isometry in the sense that

$$
\langle f, f\rangle_{L^2} = \langle \mathcal{F}(f), \mathcal{F}(f)\rangle_{\ell^2},
$$

where $\langle f, g\rangle_{L^2} = \int_{-\pi}^{\pi} f(x)g(x)dx$.

It is obvious that $\mathcal{F}$ is injective and linear. However, by the previous square mean convergence, the map $\mathcal{F}$ is far from surjective since $\mathrm{Im}(\mathcal{F})$ obvious contains the Fourier coefficient sequences of non continuous Riemann integrable functions. Therefore, we can at least naturally extend the domain of $\mathcal{F}$ to the space $R[-\pi,\pi]$. Notice that this extended map is no longer injective since two Riemann integrable functions which only differs at finitely many points have the same Fourier coefficient sequences. The more severe issue is the "incompleteness" of $R[-\pi,\pi]$ (under the metric induced by the above L^2 inner product), while on the contrast ℓ^2 itself is complete metric space (under the ℓ^2 metric induced by ℓ^2 inner product)[4] [5]. This incompleteness is indicated by the following "example".

The space $R[-\pi,\pi]$ is like $\mathbf{Q}$ with lots of holes everywhere. To complete it, i.e. to "fill in the holes", beside the above function which is discontinuous on a set of "length" nonzero, we need also to include function which is singular but square integrable, for example $\log|x|$, $|x|^{-\frac{1}{4}}$ and

$$\sum_{r_n\in\mathbf{Q}\cap[-\pi,\pi]}\frac{1}{2^n}\log|x-r_n|.$$

It is difficult to enumerate all cases in the framework of Riemann's integral, and mathematicians struggles more than half a century on this issue. It finally turns out the space (usually denoted as $L^2[-\pi,\pi]$) of *Lebesgue square integrable functions* is the "correct completion", and the above map $\mathcal{F}$ extends to this set as a bijective isometry to ℓ^2.

11 Fourier series and Fourier transform

11.1 Fourier series in complex form

Recall the Euler identity

$$e^{i\theta}=\cos\theta+i\sin\theta. \tag{11.1}$$

Suppose f defined on $[-\pi,\pi]$ can be expanded as Fourier series (for instance satisfying the condition of Dirichlet Theorem), then

$$\begin{aligned}
f(x)&=\frac{a_0}{2}+\sum_{n=1}^{\infty}(a_n\cos nx+b_n\sin nx)\\
&=\frac{a_0}{2}+\sum_{n=1}^{\infty}\left(a_n\frac{e^{inx}+e^{-inx}}{2}-ib_n\frac{e^{inx}-e^{-inx}}{2}\right)\\
&=\frac{a_0}{2}+\sum_{n=1}^{\infty}\left(\frac{a_n-ib_n}{2}e^{inx}+\frac{a_n+ib_n}{2}e^{-inx}\right)
\end{aligned} \tag{11.2}$$

Notice that

$$\begin{aligned}
\frac{a_n-ib_n}{2}&=\frac{1}{2\pi}\int_{-\pi}^{\pi}f(t)\left(\cos nt-i\sin nt\right)=\frac{1}{2\pi}\int_{-\pi}^{\pi}f(t)e^{-int}dt\\
\frac{a_n+ib_n}{2}&=\frac{1}{2\pi}\int_{-\pi}^{\pi}f(t)\left(\cos nt+i\sin nt\right)dt=\frac{1}{2\pi}\int_{-\pi}^{\pi}f(t)e^{int}dt.
\end{aligned}$$

[4]This of course needs a proof.

[5]This two problems can be analogous to the situation when we go beyond rational numbers: on the one hand, $1.00\cdots$ and $0.99\cdots$ represents the same quantity, on the other hand, non recurrent decimals are not (rational) numbers.

If we denote for $n \in \mathbf{Z}$:

$$\widehat{f}(n) = \int_{-\pi}^{\pi} f(t)e^{-int}dt, \tag{11.3}$$

then the Fourier series (11.2) takes the form

$$f(x) = \frac{1}{2\pi}\sum_{n\in\mathbf{Z}} \widehat{f}(n)e^{inx}, \tag{11.4}$$

which we called the *complex form* of Fourier series. More general, if f is defined on the interval $[-l, l]$, we can derive as above that

$$f(x) = \frac{1}{2\pi}\sum_{n\in\mathbf{Z}} \widehat{f}(n\frac{\pi}{l})e^{in\frac{\pi}{l}x}, \tag{11.5}$$

where

$$\widehat{f}(n\frac{\pi}{l}) = \frac{\pi}{l}\int_{-l}^{l} f(t)e^{-in\frac{\pi}{l}t}dt. \tag{11.6}$$

Notice that this formula makes perfect sense for f valued in $\mathbf{C}$, i.e. we are actually considering complex-valued periodic function. The function f is defined on $\mathbf{R}$ and has period $2l$, and the function $\widehat{f}$ is defined on $\frac{\pi}{l}\mathbf{Z}$ (and valued in $\mathbf{C}$), which is called the Fourier series or *discrete Fourier transform* of f. For a periodic wave f of frequency $\omega = \frac{\pi}{l}$, $\widehat{f}(n\omega)$ measures the amplitude of the frequency $n\omega$-component of f. This is a prototype of the wave-particle duality of light? 波粒二象性.

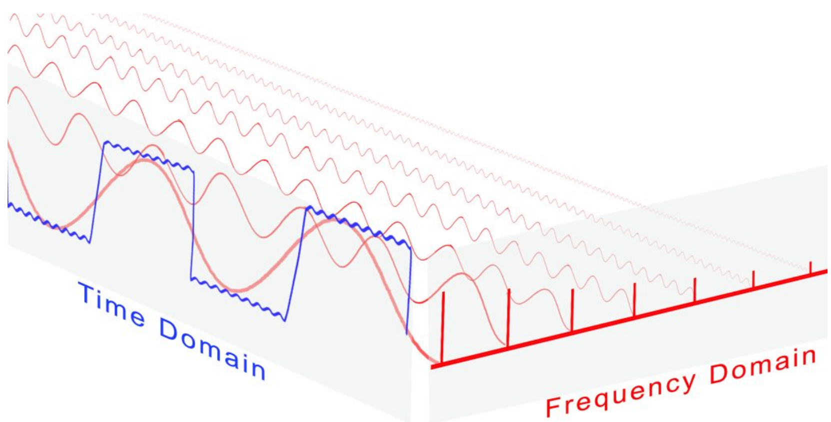


Figure 25: Time domain v.s. Frequency domain representation of signal

In signal processing, the function f is called the signal on the *time domain* and $\widehat{f}$ is called the signal on the *frequency domain*. They are two facets of the underlying signal/information, and *Fourier series/expansion* provides a duality between signal on the time domain with the corresponding signal on the frequency domain.

11.2 Properties of Fourier series

Let f, g be periodic on $\mathbf{R}$ with period 2π. Assume $f, g \in R[-\pi, \pi]$, then we can define

$$f * g(x) = \int_{-\pi}^{\pi} f(t)g(x-t)dt.$$

This is called the *convolution* of f and g. Intuitively, this represents the "weighted average of g using the weight determined by f". This is again periodic on $\mathbf{R}$ with period 2π. Recall the fact that

$$S_N(f)(x) = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x-t)\mathcal{D}_N(t)dt = f * \left(\frac{1}{\pi}\mathcal{D}_N\right)(x).$$

Theorem 11.1. *Let f be periodic on $\mathbf{R}$ with period 2π, then*

1. *(Smoothness and Decaying rate correspondence) If f is $f \in C^k$ (with $k \geq 0$), then $a_n, b_n = o\left(\frac{1}{n^k}\right)$;*

2. *(Differentiation and Magnifying correspondence) If $f \in C^1$, then $\widehat{f'}(n) = in\widehat{f}(n)$ for $n \in \mathbf{Z}$;*

3. *(Fourier inversion formula) If f is C^0 and piecewisely C^1, then*

$$f(x) = \frac{1}{2\pi}\sum_{n\in\mathbf{Z}} \widehat{f}(n)e^{inx} \quad ;$$

4. *(Shift and Phase rotation correspondence) Let $f_h(x) = f(x+h)$ be a left-shift by h of f, then $\widehat{f_h}(n) = e^{inh}\widehat{f}(n)$;*

5. *(Convolution and Multiplication correspondence) $\widehat{f * g}(n) = \widehat{f}(n)\cdot\widehat{g}(n)$ for $n \in \mathbf{Z}$.*

11.3 Real world applications: string music

Fourier series really appears at lots of places in our real world. We only look very briefly into the real world example of "vibrating strings" which is the basics of string music:

A straight string putting on the interval $[0, l]$ with two endpoints fixed vibrates according to the *Wave Equation*. The (vertical) displacement at the point $x \in [0, l]$ and time $t \in \mathbf{R}$ according to the following PDE with initial and boundary value:

$$\begin{aligned} &u_{tt} = u_{xx} \\ &u(0,t) = u(l,t) = 0 \\ &u(x,0) = f(x). \end{aligned} \tag{11.7}$$

Similarly to what we did for heat equation, we can deduce (formally) the solution:

$$u(x,t) = \sum_{n=1}^{\infty} A_n \cos \frac{m\pi}{l} t \cdot \sin \frac{m\pi}{l} x, \tag{11.8}$$

where

$$\sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{l} x = f(x)$$

is the Fourier expansion of the initial displacement function $f(x)$. Taking the following (already computed) example:

$$f(x) = \begin{cases} x, & 0 \le x \le \frac{l}{2} \\ l - x, & \frac{l}{2} \le x \le l. \end{cases}$$

Using sine series expansion, we know that

$$f(x) = \frac{4l}{\pi^2} \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{(2k-1)^2} \sin \frac{(2k-1)\pi}{l} x.$$

Therefore, the solution for plucking the string (such as guitar) from the middle point with an displacement E is

$$u(x,t) = \frac{8E}{\pi^2} \sum_{k=1}^{\infty} \frac{(-1)^k}{(2k-1)^2} \cos \frac{(2k-1)\pi}{l} t \cdot \sin \frac{(2k-1)\pi}{l} x. \tag{11.9}$$

Physically, this represents the superposition of infinitely many *standing waves* (驻波). The term corresponding to $k = 1$ is called the *fundamental tone* (基音) with frequency $\frac{\pi}{l}$, and the term corresponding to the component of frequencies $\frac{2\pi}{l}, \frac{3\pi}{l}, \cdots$ are called *overtones* (泛音). This particular example illustrates that in general the magnitude of overtones die off quite quickly with the increasing of frequency.

In general, string musical instruments are based on based on *twelve tone system* 十二平均律. We fix a base frequency ω_0 (in general taken to be 440 Hz, which corresponds to (low) $A : La$). Let $\lambda = \sqrt[12]{2}$. The following table shows the physical frequency (as mechanical sound wave in Hz.):

Common	:	$\cdots$	Fa	So	La	Si	Do	Re	Mi	Fa	So	La	Si	$\cdots$
Music note	:	$\cdots$	$\underset{\cdot}{F}$	$\underset{\cdot}{G}$	A	B	C	D	E	F	G	$\dot{A}$	$\dot{B}$	$\cdots$
frequency/ω_0	:	$\cdots$	λ^{-4}	λ^{-2}	1	λ^2	λ^3	λ^5	λ^7	λ^8	λ^{10}	2	$2\lambda^2$	$\cdots$

The twelve-tone system can produce signal which is superposition of sound of mixed frequencies. Take one octave as example:

$$T(t) = \sum_{k=0}^{11} \sum_{n=1}^{\infty} b_{k,n} \cos \left(n \cdot \lambda^k \omega_0 \cdot t + \theta_{k,n} \right) \tag{11.10}$$

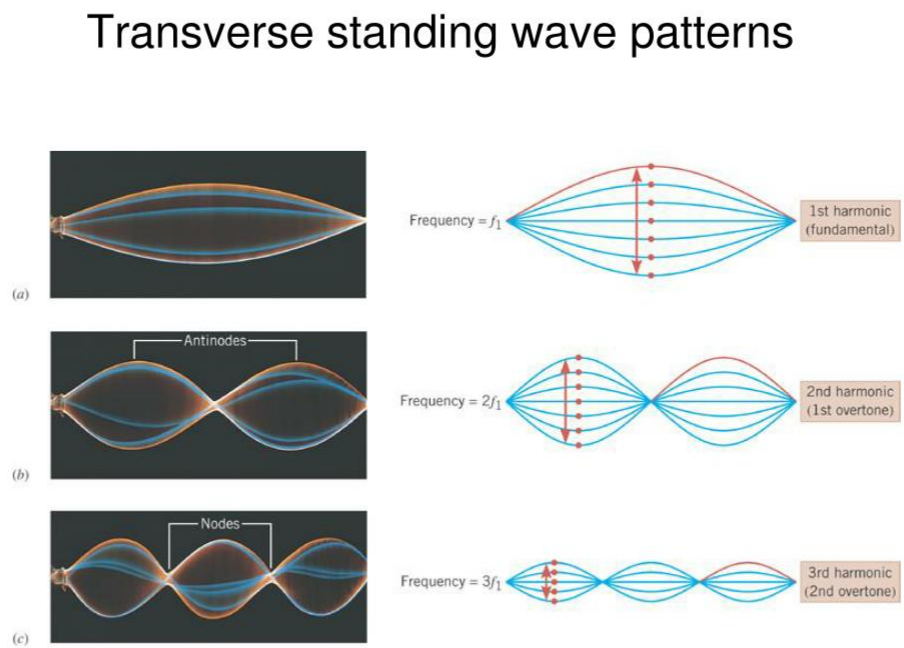


Figure 26: Sound as superposition of different frequencies harmonics

This (very rough formula) represents the magnitude of the vibration of air at a particular space point along with the flow of time. In general, for each fixed k, only the first several terms of $\{b_{k,n}\}_{n=1,2,3,\cdots}$ are significantly large enough to be counted. For instance in piano, basically only $b_{k,1}$ is nonzero, that is why we need the key (corresponding to $\dot{A}$) to generate $\sin(2\omega_0 t)$ instead of relying on A to generate it.

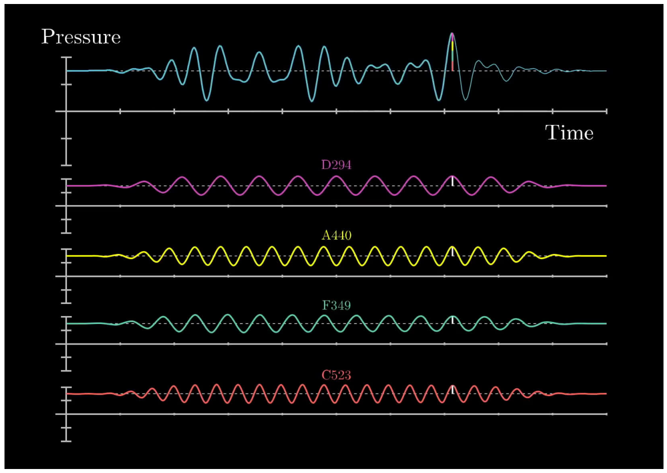


Figure 27: Sound as superposition of different frequencies

11.4 Fourier transform: a gentle introduction

Suppose f has period l, then we have

$$\begin{aligned} f(x) &= \frac{1}{2\pi}\sum_{n\in\mathbf{Z}} \widehat{f}(n\frac{\pi}{l})e^{in\frac{\pi}{l}x} \\ &= \frac{1}{2\pi}\sum_{n\in\mathbf{Z}} \frac{\pi}{l}\int_{-l}^{l} f(t)e^{-in\frac{\pi}{l}t}dt\; e^{in\frac{\pi}{l}x} \\ &= \frac{1}{2\pi}\sum_{n\in\mathbf{Z}} \left(\int_{-l}^{l} f(t)e^{-i\xi_n t}dt\right)\; e^{i\xi_n x}\cdot\Delta\xi_n \\ &= \frac{1}{2\pi}\sum_{n\in\mathbf{Z}} H_l(\xi_n)e^{i\xi_n x}\Delta\xi_n, \end{aligned}$$

which is formally a "Riemann sum" (where the sampling points are $\xi_n = n\frac{\pi}{l}$), where the function $H_l(\xi) = \int_{-l}^{l} f(t)e^{-i\xi t}dt$ is defined on $\frac{\pi}{l}\mathbf{Z}$.

As the period l gets bigger, the *supporting frequency set* $\frac{\pi}{l}\mathbf{Z}$ gets more and more dense. Intuitively, for a (not necessarily periodic) function on $\mathbf{R}$, we can replace the discrete summation (kind of Riemann sum) by continuous integration, and obtain that

$$f(x) \sim \frac{1}{2\pi}\int_{-\infty}^{\infty}\left(\int_{-\infty}^{\infty} f(t)e^{-i\xi t}dt\right)e^{i\xi x}d\xi. \tag{11.11}$$

Definition 11.2 (Fourier transform)**.** Let $f : \mathbf{R} \longrightarrow \mathbf{C}$ such that $f \in R[a,b]$

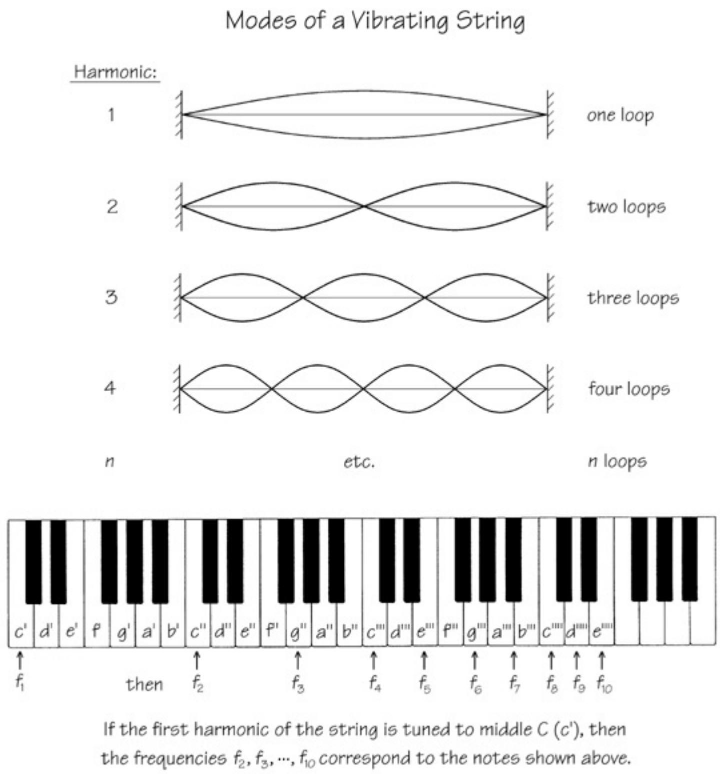


Figure 28: The relation between different notes: $f_2 = 2f_1, f_4 = 2^2 f_1, f_8 = 2^3 f_1$ and $f_3 = 2^{\frac{7}{12}} f_2 \approx 3f_1$ etc.

for any $[a, b]$ and $\int_{-\infty}^{\infty} |f(x)|dx < +\infty$. Then, the (well-defined) function

$$\widehat{f} : \mathbf{R} \longrightarrow \mathbf{C},$$
$$\xi \mapsto \int_{-\infty}^{\infty} f(t)e^{-i\xi t}dt,$$

is called the *Fourier transform* of f.

Theorem 11.3 (Fourier inversion formula)**.** *Suppose f is defined on $\mathbf{R}$ is piecewisely differentiable on any finite interval, and absolutely integrable on $\mathbf{R}$, then for all $x \in \mathbf{R}$, we have*

$$\frac{1}{2\pi}\int_{-\infty}^{\infty}\left(\int_{-\infty}^{\infty} f(t)e^{-i\xi t}dt\right)e^{i\xi x}d\xi = \frac{f(x+0)+f(x-0)}{2}.$$

In particular, if moreover f is continuous, then

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\left(\int_{-\infty}^{\infty} f(t)e^{-i\xi t}dt\right)e^{i\xi x}d\xi = \frac{1}{2\pi}\int_{-\infty}^{\infty}\widehat{f}(\xi)e^{i\xi x}d\xi.$$

The type of integral appears in Fourier transform and Fourier inversion formula is called *integral with parameter* and *improper integral with parameter*, which are the main content of next chapter.

Example 11.4.

$$f_h(x) = \begin{cases} \frac{1}{2h}, & |x| \le h, \\ 0, & |x| > h \end{cases}$$

Solution.

$$\widehat{f_h}(\xi) = \int_{-\infty}^{\infty} f(x)e^{-i\xi x}dx = \frac{1}{2h}\int_{-h}^{h} e^{-i\xi x}dx = \frac{\sin \xi x}{\xi h}|_{x=0}^{x=h} = \frac{\sin \xi h}{\xi h}, \ \ \xi \neq 0,$$
$$\widehat{f_h}(0) = 1, \ \ \xi = 0.$$

Please compare this result with the periodic square wave we've considered in Fourier series. This can also be interpreted as "to produce a spatially localized wave packet, we need a set of harmonics with continuum of frequencies".

It is more clear from this example, "the more concentrated is f in the time domain, the more sparse is $\widehat{f}$ in the frequency domain".